

DEF CON 29

Tamas K Lengyel

Fuzzing Linux with Xen



don't microwave your pets

Performance varies by use, configuration and other factors. Learn more at www.Intel.com/PerformanceIndex.

Performance results are based on testing as of dates shown in configurations and may not reflect all publicly available updates. See backup for configuration details. No product or component can be absolutely secure.

No license (express or implied, by estoppel or otherwise) to any intellectual property rights is granted by this document.

Intel does not control or audit third-party data. You should consult other sources to evaluate accuracy.

Intel disclaims all express and implied warranties, including without limitation, the implied warranties of merchantability, fitness for a particular purpose, and non-infringement, as well as any warranty arising from course of performance, course of dealing, or usage in trade.

Your costs and results may vary.

Intel technologies may require enabled hardware, software or service activation.

© Intel Corporation. Intel, the Intel logo, and other Intel marks are trademarks of Intel Corporation or its subsidiaries. Other names and brands may be claimed as the property of others.

whoami



./ **DRAKVUF®**

Black-box Binary Analysis System

LibVMI
Virtual Machine Introspection
Fast, Portable, Simple

intel®



We fuzzed the device-facing input points of several Linux drivers

We built new tools to get it done

We open-sourced them

We found a bunch of bugs & fixed them

Showing you how it was done

feedback fuzzers

Not “just” about feeding random input to your target

Feedback fuzzers monitor the target execution

Collect execution log (aka. coverage)

Compare execution from run-to-run

New code discovered? Focus on that input!

feedback fuzzers

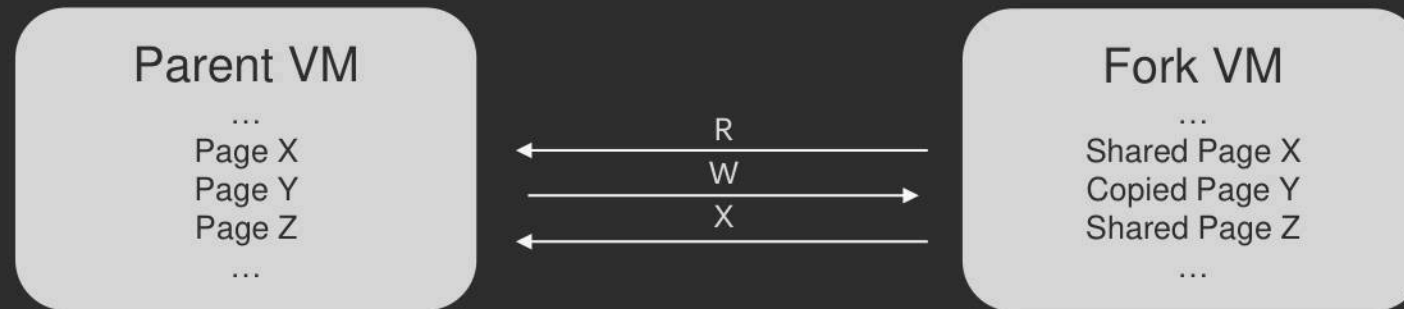
Need determinism

If your target code behaves differently between executions not due to the fuzzer, the feedback will be just noise



xen vm forking

Add determinism to kernel-code execution



Reset vCPU registers and free copied pages for better speed

vm forking speed

VM fork creation time:

$\sim 745 \mu\text{s} \sim 1300 \text{ VM/s}$

VM fork reset time:

$\sim 111 \mu\text{s} \sim 9000 \text{ reset/s}$

Measured on i5-8350U

Xen VM introspection is natively supported

- Read/write/translate guest memory

Pause vCPU & get notification in dom0 of various hw events

- CPUID
- Int3
- MTF
- EPT faults
- register writes (mov-to-cr, MSRs)
- etc.

xen vmtrace

Upstreamed in Xen 4.15

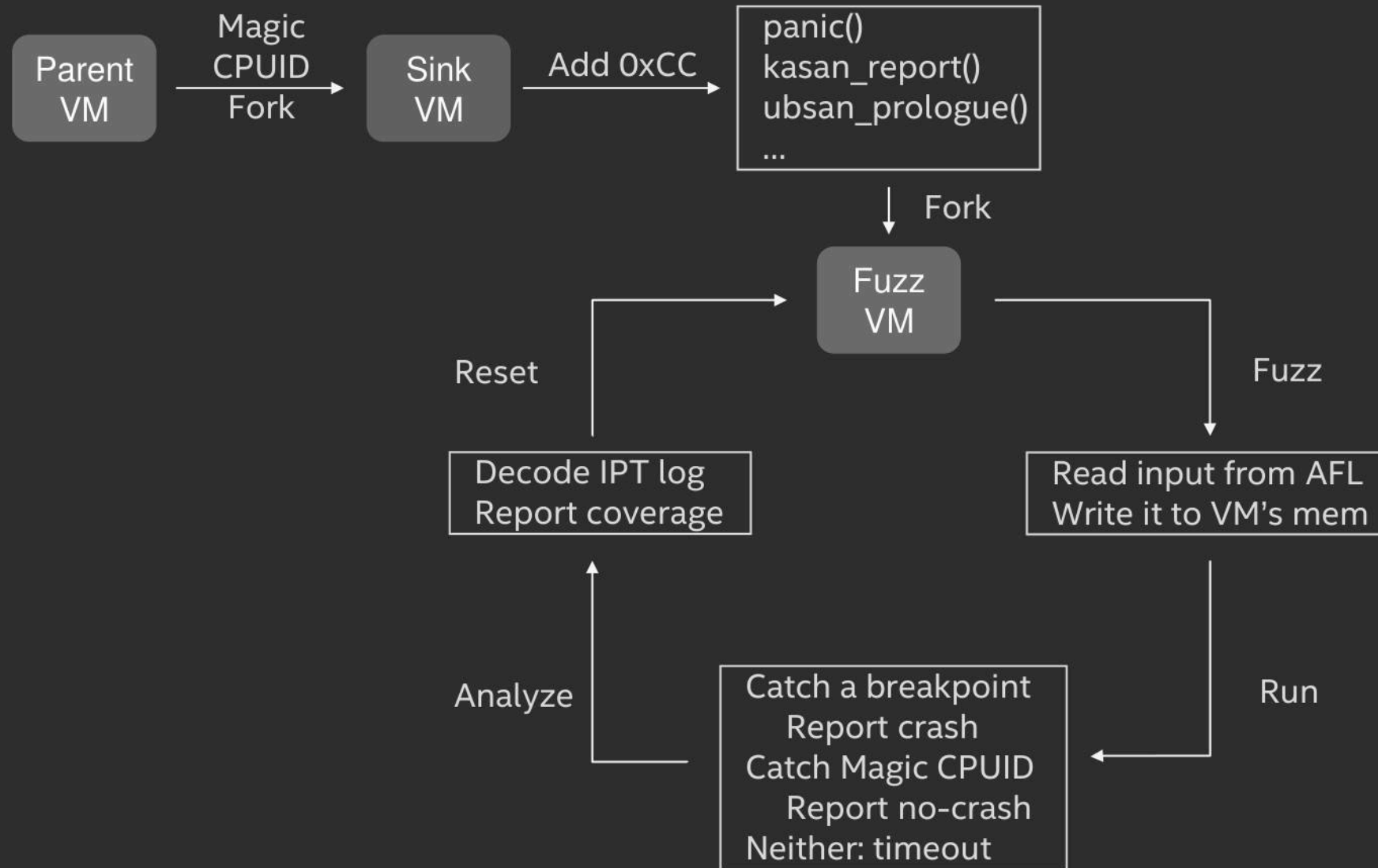
- In collaboration with CERT.pl & Citrix

Use Intel® Processor Trace to record VM execution

- CPU itself stores info in memory about execution, low overhead
- Stores only enough detail to allow for post-execution reconstruction

Can record execution of the whole VM

fuzzing on xen



```

[ OK ] Mounted Mount unit for go, revision 7736.
[ OK ] Mounted Mount unit for gtk-common-themes, revision 1514.
[ OK ] Started udev Kernel Device Manager.
[ OK ] Mounted Mount unit for gtk-common-themes, revision 1515.
[ OK ] Mounted Mount unit for snap-store, revision 518.
[ OK ] Mounted Mount unit for snap-store, revision 542.
[ OK ] Mounted Mount unit for snapd, revision 11841.
[ OK ] Mounted Mount unit for snapd, revision 12057.
[ OK ] Started Dispatch Password ...ts to Console Directory Watch.
[ OK ] Reached target Local Encrypted Volumes.
[ OK ] Found device /dev/ttyS0.
[ OK ] Listening on Load/Save RF ...itch Status /dev/rfkill Watch.
[ OK ] Found device /dev/hvc0.
[ OK ] Found device /dev/disk/by-uuid/66A8-01DE.
      Starting File System Check.../dev/disk/by-uuid/66A8-01DE...
[ OK ] Started File System Check Daemon to report status.
[ OK ] Finished File System Check...n /dev/disk/by-uuid/66A8-01DE.
      Mounting /boot/efi...
[ OK ] Mounted /boot/efi.
[ OK ] Reached target Local File Systems.
      Starting Load AppArmor profiles...
      Starting Set console font and keymap...
      Starting Tell Plymouth To Write Out Runtime Data...
      Starting Create Volatile Files and Directories...
[ OK ] Finished Set console font and keymap.
[ OK ] Finished Tell Plymouth To Write Out Runtime Data.
[ OK ] Finished Create Volatile Files and Directories.
      Starting Network Name Resolution...
      Starting Network Time Synchronization...
      Starting Update UTMP about System Boot/Shutdown...
[ OK ] Finished Update UTMP about System Boot/Shutdown.
[ OK ] Started Network Time Synchronization.
[ OK ] Reached target System Time Set.
[ OK ] Reached target System Time Synchronized.
[ OK ] Started Network Name Resolution.
[ OK ] Reached target Host and Network Name Lookups.
[ OK ] Finished Load AppArmor profiles.
      Starting Load AppArmor pro...managed internally by snapd...

```

```
root@t2:/shared/xhci_demo# cat ubuntu-20.04.cfg
```

```

arch = 'x86_64'
name = "ubuntu-20.04"
maxmem = 1024
memory = 1024
vcpus = 1
maxcpus = 1
nomigrate = 1
boot = "cd"

```

```

type = "hvm"
serial = "pty"

```

```

vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']
disk = ['phy:/dev/t2vg/ubuntu-20.04,xvda,w']

```

```
vmtrace_buf_kb=65536
```

```

device_model_args = [
    "-device", "qemu-xhci,id=qemu-xhci,p3=8",
    "-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",
    "-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",
]

```

```
vga="none"
```

```
vnc=0
```

```

root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c
Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)
Waiting for harness start (cpuid with leaf 0x13371300)

```


Disk /dev/loop5: 90.6 MiB, 94986240 bytes, 185520 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop6: 64.79 MiB, 67915776 bytes, 132648 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop7: 65.1 MiB, 68259840 bytes, 133320 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/xvda: 40 GiB, 42949672960 bytes, 83886080 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17cc52b9

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/xvda1		2048	1050623	1048576	512M	b	W95 FAT32
/dev/xvda2		1052670	83884031	82831362	39.5G	5	Extended
/dev/xvda5		1052672	83884031	82831360	39.5G	83	Linux

```
root@t2:/shared/xhci_demo#  
root@t2:/shared/xhci_demo#  
root@t2:/shared/xhci_demo# xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8192	8	r-----	2472.7
ubuntu-20.04	12	1024	1	--p----	31.4

```
root@t2:/shared/xhci_demo# cat ubuntu-20.04.cfg
```

```
arch = 'x86_64'  
name = "ubuntu-20.04"  
maxmem = 1024  
memory = 1024  
vcpus = 1  
maxcpus = 1  
nomigrate = 1  
boot = "cd"
```

```
type = "hvm"  
serial = "pty"
```

```
vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']  
disk = ['phy:/dev/t2vg/ubuntu-20.04,xvda,w']
```

```
vmtrace_buf_kb=65536
```

```
device_model_args = [  
    "-device", "qemu-xhci,id=qemu-xhci,p3=8",  
    "-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",  
    "-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",  
]
```

```
vga="none"  
vnc=0
```

```
root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c  
Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)  
Waiting for harness start (cpuid with leaf 0x13371300)  
Got start cpuid callback with leaf: 0x13371300 subleaf: 0x10  
Target buffer & size: 0xffff8880338ad9d0 16  
Parent is ready  
root@t2:/shared/xhci_demo#
```

Disk /dev/loop5: 90.6 MiB, 94986240 bytes, 185520 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop6: 64.79 MiB, 67915776 bytes, 132648 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop7: 65.1 MiB, 68259840 bytes, 133320 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/xvda: 40 GiB, 42949672960 bytes, 83886080 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17cc52b9

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/xvda1		2048	1050623	1048576	512M	b	W95 FAT32
/dev/xvda2		1052670	83884031	82831362	39.5G	5	Extended
/dev/xvda5		1052672	83884031	82831360	39.5G	83	Linux

```
root@t2:/shared/xhci_demo#  
root@t2:/shared/xhci_demo#  
root@t2:/shared/xhci_demo# xl list  
Name ID Mem VCPUs State Time(s)  
Domain-0 0 8192 8 r----- 2472.7  
ubuntu-20.04 12 1024 1 --p--- 31.4  
root@t2:/shared/xhci_demo#
```

```
root@t2:/shared/xhci_demo# cat ubuntu-20.04.cfg  
arch = 'x86_64'  
name = "ubuntu-20.04"  
maxmem = 1024  
memory = 1024  
vcpus = 1  
maxcpus = 1  
nomigrate = 1  
boot = "cd"
```

```
type = "hvm"  
serial = "pty"
```

```
vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']  
disk = ['phy:/dev/t2vg/ubuntu-20.04,xvda,w']
```

```
vmtrace_buf_kb=65536
```

```
device_model_args = [  
    "-device", "qemu-xhci,id=qemu-xhci,p3=8",  
    "-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",  
    "-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",  
]
```

```
vga="none"  
vnc=0
```

```
root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c  
Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)
```

```
Waiting for harness start (cpuid with leaf 0x13371300)  
Got start cpuid callback with leaf: 0x13371300 subleaf: 0x10  
Target buffer & size: 0xfffff8880338ad9d0 16  
Parent is ready
```

```
root@t2:/shared/xhci_demo# mkdir input  
root@t2:/shared/xhci_demo# mkdir output  
root@t2:/shared/xhci_demo# rwmem --domid 12 --read 0xfffff8880338ad9d0 --file input/seed --limit 16  
Init vmi, init_events: 0 init_paging 1 domain (null) domid 12 json (null)  
Read operation success: 16 bytes from 0xfffff8880338ad9d0  
root@t2:/shared/xhci_demo#
```


Disk /dev/loop5: 90.6 MiB, 94986240 bytes, 185520 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop6: 64.79 MiB, 67915776 bytes, 132648 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop7: 65.1 MiB, 68259840 bytes, 133320 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/xvda: 40 GiB, 42949672960 bytes, 83886080 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17cc52b9

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/xvda1		2048	1050623	1048576	512M	b	W95 FAT32
/dev/xvda2		1052670	83884031	82831362	39.5G	5	Extended
/dev/xvda5		1052672	83884031	82831360	39.5G	83	Linux

root@t2:/shared/xhci_demo#

root@t2:/shared/xhci_demo#

root@t2:/shared/xhci_demo# xl list

Name	ID	Mem	VCPU	State	Time(s)
Domain-0	0	8192	8	r-----	2472.7
ubuntu-20.04	12	1024	1	--p---	31.4

root@t2:/shared/xhci_demo# afl-fuzz -i input -o output -- kfx --domid 12 --json kernel.json --
address 0xffff8880338ad9d0 --input @@ --input-limit 16 --ptcov

root@t2:/shared/xhci_demo# cat ubuntu-20.04.cfg

```
arch = 'x86_64'  
name = "ubuntu-20.04"  
maxmem = 1024  
memory = 1024  
vcpus = 1  
maxcpus = 1  
nomigrate = 1  
boot = "cd"
```

```
type = "hvm"  
serial = "pty"
```

```
vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']  
disk = ['phy:/dev/t2vg/ubuntu-20.04,xvda,w']
```

vmtrace_buf_kb=65536

```
device_model_args = [  
    "-device", "qemu-xhci,id=qemu-xhci,p3=8",  
    "-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",  
    "-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",  
]
```

vga="none"

vnc=0

root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c

Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)

Waiting for harness start (cpuid with leaf 0x13371300)

Got start cpuid callback with leaf: 0x13371300 subleaf: 0x10

Target buffer & size: 0xffff8880338ad9d0 16

Parent is ready

root@t2:/shared/xhci_demo# mkdir input

root@t2:/shared/xhci_demo# mkdir output

root@t2:/shared/xhci_demo# rwmem --domid 12 --read 0xffff8880338ad9d0 --file input/seed --limit 16

Init vmi, init_events: 0 init_paging 1 domain (null) domid 12 json (null)

Read operation success: 16 bytes from 0xffff8880338ad9d0

root@t2:/shared/xhci_demo#

american fuzzy lop ++3.10c (default) [fast] {0}

process timing

run time : 0 days, 0 hrs, 0 min, 10 sec
last new path : 0 days, 0 hrs, 0 min, 1 sec
last uniq crash : 0 days, 0 hrs, 0 min, 4 sec
last uniq hang : none seen yet

cycle progress

now processing : 26.0 (53.1%)
paths timed out : 0 (0.00%)

stage progress

now trying : splice 10
stage execs : 11/12 (91.67%)
total execs : 16.2k
exec speed : 1711/sec

fuzzing strategy yields

bit flips : n/a, n/a, n/a
byte flips : n/a, n/a, n/a
arithmetics : n/a, n/a, n/a
known ints : n/a, n/a, n/a
dictionary : n/a, n/a, n/a
havoc/splice : 26/5068, 23/9112
py/custom : 0/0, 0/0
trim : 1.96%/79, n/a

overall results

cycles done : 0
total paths : 49
uniq crashes : 1
uniq hangs : 0

map coverage

map density : 0.17% / 0.38%
count coverage : 4.93 bits/tuple

findings in depth

avored paths : 31 (63.27%)
new edges on : 44 (89.80%)
total crashes : 2 (1 unique)
total tmouts : 0 (0 unique)

path geometry

levels : 5
pending : 21
pend fav : 5
own finds : 48
imported : 0
stability : 45.14%

[cpu000: 25%]

^A

root@t2:/shared/xhci_demo# cat ubuntu-20.04.cfg

arch = 'x86_64'
name = "ubuntu-20.04"
maxmem = 1024
memory = 1024
vcpus = 1
maxcpus = 1
nomigrate = 1
boot = "cd"

type = "hvm"
serial = "pty"

vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']
disk = ['phy:/dev/t2vg/ubuntu-20.04,xvda,w']

vmtrace_buf_kb=65536

device_model_args = [
"-device", "qemu-xhci,id=qemu-xhci,p3=8",
"-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",
"-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",
]

vga="none"

vnc=0

root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c

Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)

Waiting for harness start (cpuid with leaf 0x13371300)

Got start cpuid callback with leaf: 0x13371300 subleaf: 0x10

Target buffer & size: 0xffff8880338ad9d0 16

Parent is ready

root@t2:/shared/xhci_demo# mkdir input

root@t2:/shared/xhci_demo# mkdir output

root@t2:/shared/xhci_demo# rwmem --domid 12 --read 0xffff8880338ad9d0 --file input/seed --limit 16

Init vmi, init_events: 0 init_paging 1 domain (null) domid 12 json (null)

Read operation success: 16 bytes from 0xffff8880338ad9d0

root@t2:/shared/xhci_demo#

there is more to fuzzing



Memory made accessible to a device

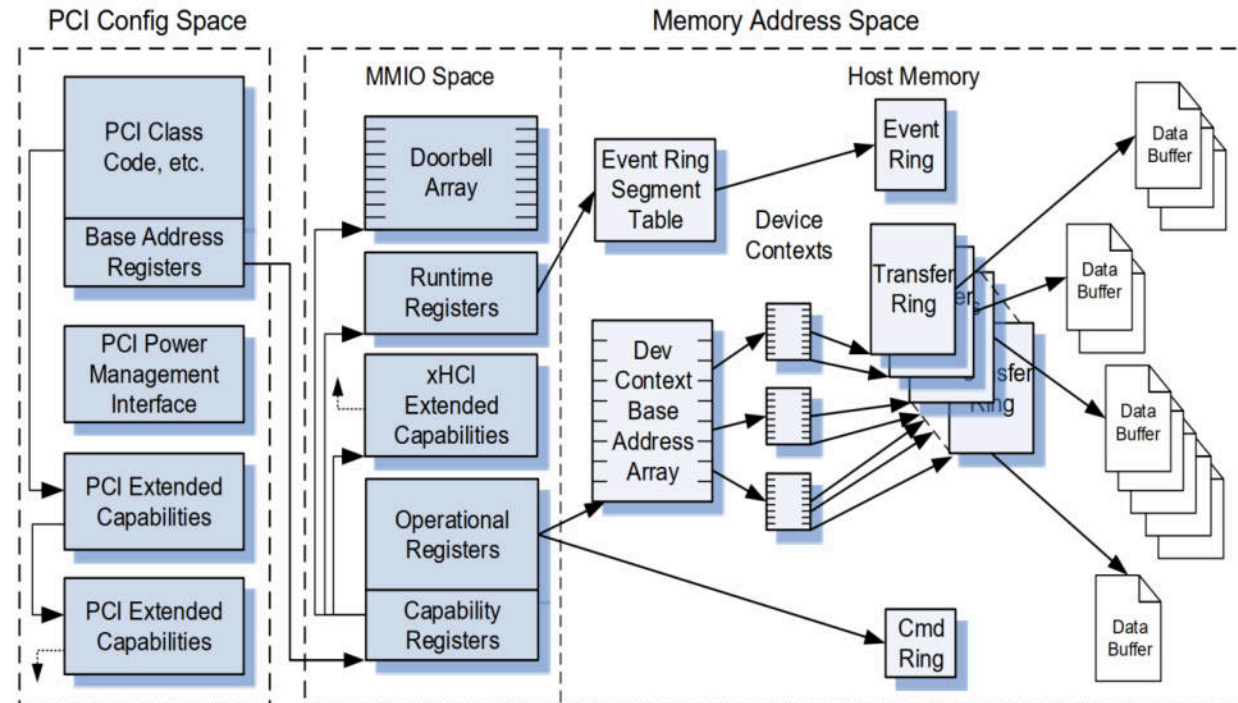
- Facilitates fast I/O operations
- IOMMU can restrict device to access only DMA pages

Finding where Linux reads from DMA isn't trivial

- `__iomem`, `be{16/32/64}_to_cpu`, `le{16/32/64}_to_cpu`, `dma_addr_t`
- Cross-reference with ftrace output

let's just read the spec

Figure 3-3: General Architecture of the eXtensible Host Controller Interface



<https://www.intel.com/content/www/us/en/products/docs/io/universal-serial-bus/extendible-host-controller-interface-usb-xhci.html>

```

diff --git a/drivers/usb/host/xhci-ring.c b/drivers/usb/host/xhci-ring.c
index 167dae117f73..ca29fb6c0b1f 100644
--- a/drivers/usb/host/xhci-ring.c
+++ b/drivers/usb/host/xhci-ring.c
@@ -2730,10 +2730,16 @@ static int xhci_handle_event(struct xhci_hcd *xhci)
    }

    event = xhci->event_ring->dequeue;
+
+    harness_start(0x13371300, event, sizeof(*event));
+
    /* Does the HC or OS own the TRB? */
    if ((le32_to_cpu(event->event_cmd.flags) & TRB_CYCLE) !=
        xhci->event_ring->cycle_state)
+
+    {
+        harness_stop();
+        return 0;
+    }

    trace_xhci_handle_event(xhci->event_ring, &event->generic);
@@ -2772,6 +2778,7 @@ static int xhci_handle_event(struct xhci_hcd *xhci)
    * to make sure a watchdog timer didn't mark the host as non-responsive.
    */
    if (xhci->xhc_state & XHCI_STATE_DYING) {
+
+        harness_stop();
+        xhci_dbg(xhci, "xHCI host dying, returning from "
+                "event handler.\n");

        return 0;
@@ -2781,6 +2788,8 @@ static int xhci_handle_event(struct xhci_hcd *xhci)
    /* Update SW event ring dequeue pointer */
    inc_deq(xhci, xhci->event_ring);

+    harness_stop();
+
    /* Are there more items on the event ring? Caller will call us again to
    * check.
    */

```



```

diff --git a/drivers/usb/host/xhci.h b/drivers/usb/host/xhci.h
index ebb359ebb261..77be2863cfd9 100644
--- a/drivers/usb/host/xhci.h
+++ b/drivers/usb/host/xhci.h
@@ -2750,4 +2750,22 @@ static inline const char *xhci_decode_ep_context(u32 info, u32 info2, u64 deq,
        return str;
    }

+static inline void harness_start(unsigned int magic, void *a, size_t s)
+{
+    asm volatile ("cpuid"
+        : "=a" (magic), "=c" (magic), "=S" (magic)
+        : "a" (magic), "c" (s), "S" (a)
+        : "bx", "dx");
+}
+
+static inline void harness_stop(void)
+{
+    unsigned int tmp;
+
+    asm volatile ("cpuid"
+        : "=a" (tmp)
+        : "a" (0x13371337)
+        : "bx", "cx", "dx");
+}
+

```

triaging

Can't just “log in and gather logs”

- No network
- No disk
- No console

The dmesg buffer of Linux is in RAM

- We can carve it out!

xen gdbserver

It's been shipping with Xen since 2009

Minimal gdb bridge

Build Linux with:

```
CONFIG_DEBUG_INFO=y  
CONFIG_GDB_SCRIPTS=y  
CONFIG_FRAME_POINTER=y  
CONFIG_UNWINDER_FRAME_POINTER=y  
CONFIG_RANDOMIZE_BASE=n
```

```

american fuzzy lop ++3.10c (default) [fast] {0}
process timing
  run time : 0 days, 0 hrs, 0 min, 12 sec
  last new path : 0 days, 0 hrs, 0 min, 2 sec
  last uniq crash : 0 days, 0 hrs, 0 min, 5 sec
  last uniq hang : none seen yet
cycle progress
  now processing : 46.2 (93.9%)
  paths timed out : 0 (0.00%)
stage progress
  now trying : splice 2
  stage execs : 39/44 (88.64%)
  total execs : 18.6k
  exec speed : 1686/sec
fuzzing strategy yields
  bit flips : n/a, n/a, n/a
  byte flips : n/a, n/a, n/a
  arithmetics : n/a, n/a, n/a
  known ints : n/a, n/a, n/a
  dictionary : n/a, n/a, n/a
  havoc/splice : 26/5938, 23/10.6k
  py/custom : 0/0, 0/0
  trim : 1.60%/98, n/a
overall results
  cycles done : 0
  total paths : 49
  uniq crashes : 1
  uniq hangs : 0
map coverage
  map density : 0.19% / 0.38%
  count coverage : 4.93 bits/tuple
findings in depth
  favored paths : 31 (63.27%)
  new edges on : 44 (89.80%)
  total crashes : 2 (1 unique)
  total tmouts : 0 (0 unique)
path geometry
  levels : 5
  pending : 16
  pend fav : 0
  own finds : 48
  imported : 0
  stability : 45.14%
[cpu000: 37%]

```

+++ Testing aborted by user +++

^C[+] We're done here. Have a nice day!

```

root@t2:/shared/xhci_demo# kfx --domid 12 --json kernel.json --address 0xffff8880338ad9d0 --in
put output/default/crashes/id\:000000\,sig\:06\,src\:000031+000016\,time\:6669\,op\:splice\,re
p\:4 --input-limit 16 --nocov --debug
##### START #####
Init vmi, init_events: 0 init_paging 0 domain (null) domid 12 json (null)
Parent is ready
Fork VMs created: 12 -> 15 -> 16
Init vmi, init_events: 0 init_paging 1 domain (null) domid 15 json kernel.json

```

```

root@t2:/shared/xhci_demo# cat ubuntu-20.04.cfg

```

```

arch = 'x86_64'
name = "ubuntu-20.04"
maxmem = 1024
memory = 1024
vcpus = 1
maxcpus = 1
nomigrate = 1
boot = "cd"

```

```

type = "hvm"
serial = "pty"

```

```

vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']
disk = ['phy:/dev/t2vg/ubuntuu-20.04,xvda,w']

```

```

vmtrace_buf_kb=65536

```

```

device_model_args = [
    "-device", "qemu-xhci,id=qemu-xhci,p3=8",
    "-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",
    "-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",
]

```

```

vga="none"

```

```

vnc=0

```

```

root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c

```

```

Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)

```

```

Waiting for harness start (cpuid with leaf 0x13371300)

```

```

Got start cpuid callback with leaf: 0x13371300 subleaf: 0x10

```

```

Target buffer & size: 0xffff8880338ad9d0 16

```

```

Parent is ready

```

```

root@t2:/shared/xhci_demo# mkdir input

```

```

root@t2:/shared/xhci_demo# mkdir output

```

```

root@t2:/shared/xhci_demo# rwmem --domid 12 --read 0xffff8880338ad9d0 --file input/seed --limi
t 16

```

```

Init vmi, init_events: 0 init_paging 1 domain (null) domid 12 json (null)

```

```

Read operation success: 16 bytes from 0xffff8880338ad9d0

```

```

root@t2:/shared/xhci_demo#

```


+++ Testing aborted by user +++

^C[+] We're done here. Have a nice day!

```
root@t2:/shared/xhci_demo# kfx --domid 12 --json kernel.json --address 0xffff8880338ad9d0 --input
output/default/crashes/id\:000000\,sig\:06\,src\:000031+000016\,time\:6669\,op\:splice\,rep\
p\:4 --input-limit 16 --nocov --debug
```

START

Init vmi, init_events: 0 init_paging 0 domain (null) domid 12 json (null)

Parent is ready

Fork VMs created: 12 -> 15 -> 16

Init vmi, init_events: 0 init_paging 1 domain (null) domid 15 json kernel.json

Creating sink list from built-in information listed in sink.h

Failed to find address for sink kmsan_report in the JSON

Setting breakpoint on sink ubsan_prologue 0xffffffff82224498 -> 0x2224498

Setting breakpoint on sink kasan_report 0xffffffff813ff330 -> 0x13ff330

Setting breakpoint on sink asm_exc_page_fault 0xffffffff82400ac0 -> 0x2400ac0

Failed to find address for sink page_fault in the JSON

Setting breakpoint on sink oops_begin 0xffffffff8104d6f0 -> 0x104d6f0

Setting breakpoint on sink panic 0xffffffff821ee84b -> 0x21ee84b

Sinks are ready

Init vmi, init_events: 1 init_paging 1 domain (null) domid 16 json (null)

Setup trace

Setup trace finished

VM Fork is ready for fuzzing

Starting fuzzer on 16

Running in standalone mode

Get 16 bytes of input from output/default/crashes/id:000000,sig:06,src:000031+000016,time:6669
,op:splice,rep:4

Got input size 15

Starting trace from 0xffffffff81d8739a.

Writing 15 bytes of input to 0xffff8880338ad9d0

Starting fuzz loop

[TRACER int3] RIP: 0xffffffff82224498

Sink ubsan_prologue! Tracer counter: 0. Crash: 1.

Stopping fuzz loop.

Result: crash

Closing tracer

DONE

root@t2:/shared/xhci_demo#

root@t2:/shared/xhci_demo# cat ubuntu-20.04.cfg

arch = 'x86_64'

name = "ubuntu-20.04"

maxmem = 1024

memory = 1024

vcpus = 1

maxcpus = 1

nomigrate = 1

boot = "cd"

type = "hvm"

serial = "pty"

vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']

disk = ['phy:/dev/t2vg/ubuntu-20.04,xvda,w']

vmtrace_buf_kb=65536

device_model_args = [

"-device", "qemu-xhci,id=qemu-xhci,p3=8",

"-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",

"-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",

]

vga="none"

vnc=0

root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c

Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)

Waiting for harness start (cpuid with leaf 0x13371300)

Got start cpuid callback with leaf: 0x13371300 subleaf: 0x10

Target buffer & size: 0xffff8880338ad9d0 16

Parent is ready

root@t2:/shared/xhci_demo# mkdir input

root@t2:/shared/xhci_demo# mkdir output

root@t2:/shared/xhci_demo# rwmem --domid 12 --read 0xffff8880338ad9d0 --file input/seed --limit 16

Init vmi, init_events: 0 init_paging 1 domain (null) domid 12 json (null)

Read operation success: 16 bytes from 0xffff8880338ad9d0

root@t2:/shared/xhci_demo#


```

Got input size 15
Starting trace from 0xffffffff81d8739a.
Writing 15 bytes of input to 0xffff8880338ad9d0
Starting fuzz loop
[TRACER int3] RIP: 0xffffffff82224498
Sink ubsan_prologue! Tracer counter: 0. Crash: 1.
Stopping fuzz loop.
Result: crash
Closing tracer
##### DONE #####
root@t2:/shared/xhci_demo# kfx --domid 12 --json kernel.json --address 0xffff8880338ad9d0 --input output/default/crashes/id\:\:000000\,sig\:\:06\,src\:\:000031+000016\,time\:\:6669\,op\:\:splice\,rep\:\:4 --input-limit 16 --nocov --debug --sink ubsan_epilogue --keep
##### START #####
Init vmi, init_events: 0 init_paging 0 domain (null) domid 12 json (null)
Parent is ready
Fork VMs created: 12 -> 17 -> 18
Init vmi, init_events: 0 init_paging 1 domain (null) domid 17 json kernel.json
Setting breakpoint on sink ubsan_epilogue 0xffffffff82224582 -> 0x2224582
Sinks are ready
Init vmi, init_events: 1 init_paging 1 domain (null) domid 18 json (null)
Setup trace
Setup trace finished
VM Fork is ready for fuzzing
Starting fuzzer on 18
Running in standalone mode
Get 16 bytes of input from output/default/crashes/id:000000,sig:06,src:000031+000016,time:6669,op:splice,rep:4
Got input size 15
Starting trace from 0xffffffff81d8739a.
Writing 15 bytes of input to 0xffff8880338ad9d0
Starting fuzz loop
[TRACER int3] RIP: 0xffffffff82224582
Sink ubsan_epilogue! Tracer counter: 0. Crash: 1.
Stopping fuzz loop.
Result: crash
Closing tracer
##### DONE #####
root@t2:/shared/xhci_demo#

```

```

root@t2:/shared/xhci_demo# cat ubuntu-20.04.cfg
arch = 'x86_64'
name = "ubuntu-20.04"
maxmem = 1024
memory = 1024
vcpus = 1
maxcpus = 1
nomigrate = 1
boot = "cd"

type = "hvm"
serial = "pty"

vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']
disk = ['phy:/dev/t2vg/ubuntu-20.04,xvda,w']

vmtrace_buf_kb=65536

device_model_args = [
    "-device", "qemu-xhci,id=qemu-xhci,p3=8",
    "-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",
    "-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",
]

vga="none"
vnc=0
root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c
Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)
Waiting for harness start (cpuid with leaf 0x13371300)
Got start cpuid callback with leaf: 0x13371300 subleaf: 0x10
Target buffer & size: 0xffff8880338ad9d0 16
Parent is ready
root@t2:/shared/xhci_demo# mkdir input
root@t2:/shared/xhci_demo# mkdir output
root@t2:/shared/xhci_demo# rwmem --domid 12 --read 0xffff8880338ad9d0 --file input/seed --limit 16
Init vmi, init_events: 0 init_paging 1 domain (null) domid 12 json (null)
Read operation success: 16 bytes from 0xffff8880338ad9d0
root@t2:/shared/xhci_demo#

```

```

Closing tracer
##### DONE #####
root@t2:/shared/xhci_demo# kfx --domid 12 --json kernel.json --address 0xffff8880338ad9d0 --in
put output/default/crashes/id\:000000\,sig\:06\,src\:000031+000016\,time\:6669\,op\:splice\,re
p\:4 --input-limit 16 --nocov --debug --sink ubsan_epilogue --keep
##### START #####
Init vmi, init_events: 0 init_paging 0 domain (null) domid 12 json (null)
Parent is ready
Fork VMs created: 12 -> 17 -> 18
Init vmi, init_events: 0 init_paging 1 domain (null) domid 17 json kernel.json
Setting breakpoint on sink ubsan_epilogue 0xffffffff82224582 -> 0x2224582
Sinks are ready
Init vmi, init_events: 1 init_paging 1 domain (null) domid 18 json (null)
Setup trace
Setup trace finished
VM Fork is ready for fuzzing
Starting fuzzer on 18
Running in standalone mode
Get 16 bytes of input from output/default/crashes/id:000000,sig:06,src:000031+000016,time:6669
,op:splice,rep:4
Got input size 15
Starting trace from 0xffffffff81d8739a.
Writing 15 bytes of input to 0xffff8880338ad9d0
Starting fuzz loop
[TRACER int3] RIP: 0xffffffff82224582
Sink ubsan_epilogue! Tracer counter: 0. Crash: 1.
Stopping fuzz loop.
Result: crash
Closing tracer
##### DONE #####
root@t2:/shared/xhci_demo# xl list

```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8192	8	r-----	2499.5
ubuntu-20.04	12	1023	1	--p---	31.4
(null)	17	0	1	--p---	0.0
(null)	18	0	1	--p---	0.0

```

root@t2:/shared/xhci_demo# gdbSX -a 18 64 4567
Listening on port 4567

```

```

arch = 'x86_64'
name = "ubuntu-20.04"
maxmem = 1024
memory = 1024
vcpus = 1
maxcpus = 1
nomigrate = 1
boot = "cd"

type = "hvm"
serial = "pty"

vif = ['bridge=xenbr0,mac=00:07:5B:BC:00:00']
disk = ['phy:/dev/t2vg/ubuntu-20.04,xvda,w']

vmtrace_buf_kb=65536

device_model_args = [
    "-device", "qemu-xhci,id=qemu-xhci,p3=8",
    "-drive", "file=/tmp/usbstick,if=none,id=usbstick,format=raw",
    "-device", "usb-storage,bus=qemu-xhci.0,drive=usbstick,port=2",
]

vga="none"
vnc=0
root@t2:/shared/xhci_demo# kfx --setup --domain ubuntu-20.04 --magic-mark 0x13371300 -c
Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)
Waiting for harness start (cpuid with leaf 0x13371300)
Got start cpuid callback with leaf: 0x13371300 subleaf: 0x10
Target buffer & size: 0xffff8880338ad9d0 16
Parent is ready
root@t2:/shared/xhci_demo# mkdir input
root@t2:/shared/xhci_demo# mkdir output
root@t2:/shared/xhci_demo# rwmem --domid 12 --read 0xffff8880338ad9d0 --file input/seed --limi
t 16
Init vmi, init_events: 0 init_paging 1 domain (null) domid 12 json (null)
Read operation success: 16 bytes from 0xffff8880338ad9d0
root@t2:/shared/xhci_demo# cd ../linux/
root@t2:/shared/linux# gdb vmlinux -ex 'target remote :4567' -ex 'lx-dmesg' -ex 'q'

```



```

put output/default/crashes/id\:000000\,sig\:06\,src\:000031+000016\,time\:6669\,op\:splice\,re
p\:4 --input-limit 16 --nocov --debug --sink ubsan_epilogue --keep
##### START #####
Init vmi, init_events: 0 init_paging 0 domain (null) domid 12 json (null)
Parent is ready
Fork VMs created: 12 -> 17 -> 18
Init vmi, init_events: 0 init_paging 1 domain (null) domid 17 json kernel.json
Setting breakpoint on sink ubsan_epilogue 0xffffffff82224582 -> 0x2224582
Sinks are ready
Init vmi, init_events: 1 init_paging 1 domain (null) domid 18 json (null)
Setup trace
Setup trace finished
VM Fork is ready for fuzzing
Starting fuzzer on 18
Running in standalone mode
Get 16 bytes of input from output/default/crashes/id:000000,sig:06,src:000031+000016,time:6669
,op:splice,rep:4
Got input size 15
Starting trace from 0xffffffff81d8739a.
Writing 15 bytes of input to 0xfffff8880338ad9d0
Starting fuzz loop
[TRACER int3] RIP: 0xffffffff82224582
Sink ubsan_epilogue! Tracer counter: 0. Crash: 1.
Stopping fuzz loop.
Result: crash
Closing tracer
##### DONE #####
root@t2:/shared/xhci_demo# xl list

```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8192	8	r-----	2499.5
ubuntu-20.04	12	1023	1	--p---	31.4
(null)	17	0	1	--p---	0.0
(null)	18	0	1	--p---	0.0

```

root@t2:/shared/xhci_demo# gdbSX -a 18 64 4567
Listening on port 4567
Remote debugging from host 127.0.0.1
WARN: k pkt:k
putpkt(read)Detaching from guest
root@t2:/shared/xhci_demo#

```

```

5.172237] lp: driver loaded but no devices found
5.191441] ppdev: user-space parallel port driver
5.196170] systemd[1]: Starting udev Coldplug all Devices...
5.227720] EXT4-fs (xvda5): re-mounted. Opts: errors=remount-ro
5.256519] systemd[1]: Starting Uncomplicated firewall...
5.290519] systemd[1]: Started Journal Service.
5.459965] Adding 1918356k swap on /swapfile. Priority:-2 extents:5 across:1951124k SSFS
5.547991] systemd-journal[202]: Received client request to flush runtime journal.
7.595898] cryptd: max_cpu_qlen set to 1000
7.611235] AVX2 version of gcm_enc/dec engaged.
7.611238] AES CTR mode by8 optimization enabled
8.068796] input: Xen Virtual Keyboard as /devices/virtual/input/input5
8.074412] input: Xen Virtual Pointer as /devices/virtual/input/input6
19.559953] audit: type=1400 audit(1623688469.128:2): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="nvidia_modprobe" pid=364 comm="apparmor_parser"
19.559969] audit: type=1400 audit(1623688469.128:3): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="nvidia_modprobe//kmod" pid=364 comm="apparmor_parser"
19.602616] audit: type=1400 audit(1623688469.168:4): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="/usr/lib/cups/backend/cups-pdf" pid=365 comm="apparmor_parser"
19.602633] audit: type=1400 audit(1623688469.168:5): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="/usr/sbin/cupsd" pid=365 comm="apparmor_parser"
19.602646] audit: type=1400 audit(1623688469.168:6): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="/usr/sbin/cupsd//third_party" pid=365 comm="apparmor_parser"
19.626879] audit: type=1400 audit(1623688469.192:7): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="lsb_release" pid=368 comm="apparmor_parser"
19.636088] audit: type=1400 audit(1623688469.204:8): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="ippusbxd" pid=369 comm="apparmor_parser"
19.656002] audit: type=1400 audit(1623688469.224:9): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="/usr/sbin/cups-browsed" pid=370 comm="apparmor_parser"
19.669014] audit: type=1400 audit(1623688469.236:10): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="/usr/bin/man" pid=371 comm="apparmor_parser"
19.669031] audit: type=1400 audit(1623688469.236:11): apparmor="STATUS" operation="profile_
load" profile="unconfined" name="man_filter" pid=371 comm="apparmor_parser"
129.841624] =====
129.841798] UBSAN: array-index-out-of-bounds in drivers/usb/host/xhci-ring.c:2368:17
129.841886] index -1 is out of range for type 'xhci_virt_ep [31]'
root@t2:/shared/linux#

```

triaging beyond the basics

Gathering the dmesg log has been sufficient to triage most bugs

Not all of them

Sink may get triggered by code far away from the driver


```

static int igb_clean_rx_irq(struct igb_q_vector *q_vector, const int budget)
{
    struct igb_ring *rx_ring = q_vector->rx.ring;
    struct sk_buff *skb = rx_ring->skb;
    unsigned int total_bytes = 0, total_packets = 0;
    u16 cleaned_count = igb_desc_unused(rx_ring);

    while (likely(total_packets < budget)) {
        union e1000_adv_rx_desc *rx_desc;
        struct igb_rx_buffer *rx_buffer;
        unsigned int size;

        harness_stop();

        /* return some buffers to hardware, one at a time is too slow */
        if (cleaned_count >= IGB_RX_BUFFER_WRITE) {
            igb_alloc_rx_buffers(rx_ring, cleaned_count);
            cleaned_count = 0;
        }

        rx_desc = IGB_RX_DESC(rx_ring, rx_ring->next_to_clean);

        harness_start(0x13371500, rx_desc, sizeof(*rx_desc));
    }
}

```



```
BUG: KASAN: null-ptr-deref in gro_pull_from_frag0+0x98/0x1f0
Read of size 28 at addr 0000000000000000 by task rcu_sched/11
```

```
CPU: 0 PID: 11 Comm: rcu_sched Not tainted 5.4.91kfx-next-2+ #7
Hardware name: Xen HVM domU, BIOS 4.14.1 12/21/2020
```

```
Call Trace:
```

```
<IRQ>
dump_stack+0x96/0xc7
__kasan_report.cold+0x65/0x92
? gro_pull_from_frag0+0x98/0x1f0
kasan_report+0x14/0x20
check_memory_region+0x129/0x1b0
memcpy+0x24/0x50
gro_pull_from_frag0+0x98/0x1f0
dev_gro_receive+0xd2/0xd10
? __asan_loadN+0xf/0x20
? skb_gro_reset_offset+0x8f/0x140
napi_gro_receive+0x56/0x1e0
igb_poll+0xc48/0x1eb0
net_rx_action+0x29c/0x700
? napi_busy_loop+0x3d0/0x3d0
```

Bug triggers in `net/core/dev.c`, NOT in IGB itself:

```
static void gro_pull_from_frag0(struct sk_buff *skb, int grow)
{
    struct skb_shared_info *pinfo = skb_shinfo(skb);

    BUG_ON(skb->end - skb->tail < grow);

    memcpy(skb_tail_pointer(skb), NAPI_GRO_CB(skb)->frag0, grow);
}
```

```
root@ccg-cr fw-bld11:/home/cros/demo# xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	24000	12	r-----	5688.7
ubuntu-2004	37	4030	1	--p---	24.7
(null)	38	0	1	--p---	0.0
(null)	39	0	1	-----	410.8

```
root@ccg-cr fw-bld11:/home/cros/demo# kfx --domid 37^C
```

```
root@ccg-cr fw-bld11:/home/cros/demo# cat setup.log
```

```
Init vmi, init_events: 1 init_paging 0 domain (null) domid 37 json (null)
```

```
Waiting for harness start (cpuid with leaf 0x13371500)
```

```
Got start cpuid callback with leaf: 0x13371500 subleaf: 0x10
```

```
Target buffer & size: 0xffff8880b7c13000 16
```

```
Parent is ready
```

```
root@ccg-cr fw-bld11:/home/cros/demo# kfx --domid 37 --json kfx-2.json --address 0xffff8880b7c13000 --input output/default/crashes/id\:000000\,sig\:06\,src\:000000\,time\:2090\,op\:havoc\,rep\:8 --input-limit 16 --nocov --debug > debug.log
```

```

root@ccg-crfw-bld11:/home/cros/demo# cat debug.log
##### START #####
Init vmi, init_events: 0 init_paging 0 domain (null) domid 37 json (null)
Parent is ready
Fork VMs created: 37 -> 58 -> 59
Init vmi, init_events: 0 init_paging 1 domain (null) domid 58 json kfx-2.json
Creating sink list from built-in information listed in sink.h
Setting breakpoint on sink ubsan_prologue 0xffffffff82502bcb -> 0x2502bcb
Setting breakpoint on sink kasan_report 0xffffffff8149e050 -> 0x149e050
Setting breakpoint on sink page_fault 0xffffffff82601250 -> 0x2601250
Setting breakpoint on sink oops_begin 0xffffffff81060560 -> 0x1060560
Setting breakpoint on sink panic 0xffffffff824d2049 -> 0x24d2049
Sinks are ready
Init vmi, init_events: 1 init_paging 1 domain (null) domid 59 json (null)
Setup trace
Setup trace finished
VM Fork is ready for fuzzing
Starting fuzzer on 59
Running in standalone mode
Get 16 bytes of input from output/default/crashes/id:000000,sig:06,src:000000,time:2090,op:havoc,rep:8
Got input size 16
Starting trace from 0xffffffff81e07f58.
Writing 16 bytes of input to 0xffff8880b7c13000
Starting fuzz loop
[TRACER cpuid] RIP: 0xffffffff81e08069
CPUID leaf 13371501
[TRACER cpuid] RIP: 0xffffffff81e08072
CPUID leaf ffff8880
[TRACER cpuid] RIP: 0xffffffff81e082c8
CPUID leaf 13371502
[TRACER cpuid] RIP: 0xffffffff81e082d0
CPUID leaf ffff8880
[TRACER cpuid] RIP: 0xffffffff81e08306
CPUID leaf 13371503
[TRACER cpuid] RIP: 0xffffffff81e0830f
CPUID leaf ffff8880
[TRACER int3] RIP: 0xffffffff8149e050
Sink kasan_report! Tracer counter: 0. Crash: 1.
Stopping fuzz loop.
Result: crash
Closing tracer
##### DONE #####
root@ccg-crfw-bld11:/home/cros/demo#

```



```
root@ccg-cr fw-bld11:/home/cros/demo# forkvm 37
Fork VM id: 60
root@ccg-cr fw-bld11:/home/cros/demo# rwmem --domid ^C
root@ccg-cr fw-bld11:/home/cros/demo# cat setup.log
Init vmi, init_events: 1 init_paging 0 domain (null) domid 37 json (null)
Waiting for harness start (cpuid with leaf 0x13371500)
Got start cpuid callback with leaf: 0x13371500 subleaf: 0x10
Target buffer & size: 0xffff8880b7c13000 16
Parent is ready
root@ccg-cr fw-bld11:/home/cros/demo# rwmem --domid 60 --write 0xffff8880b7c13000 --input output/default/crashes/id\:000000\,sig\:06\,src\:000000\,time\:2090\,op\:havoc\,rep\
:8 --limit 16
rwmem: unrecognized option '--input'
Usage:
    --domid <domid>
    --read <address>
    --write <address>
    --file <input/output file>
    --limit <input/output limit>

Optional:
    --pagetable <address> (-1 for physical memory access)
root@ccg-cr fw-bld11:/home/cros/demo# rwmem --domid 60 --write 0xffff8880b7c13000 --file output/default/crashes/id\:000000\,sig\:06\,src\:000000\,time\:2090\,op\:havoc\,rep\
:8 --limit 16
Init vmi, init_events: 0 init_paging 1 domain (null) domid 60 json (null)
Write operation success: 16 bytes to 0xffff8880b7c13000
root@ccg-cr fw-bld11:/home/cros/demo#
```

```

##### START #####
Init vmi, init_events: 0 init_paging 0 domain (null) domid 37 json (null)
Parent is ready
Fork VMs created: 37 -> 58 -> 59
Init vmi, init_events: 0 init_paging 1 domain (null) domid 58 json kfx-2.json
Creating sink list from built-in information listed in sink.h
Setting breakpoint on sink ubsan_prologue 0xffffffff82502bcb -> 0x2502bcb
Setting breakpoint on sink kasan_report 0xffffffff8149e050 -> 0x149e050
Setting breakpoint on sink page_fault 0xffffffff82601250 -> 0x2601250
Setting breakpoint on sink oops_begin 0xffffffff81060560 -> 0x1060560
Setting breakpoint on sink panic 0xffffffff824d2049 -> 0x24d2049
Sinks are ready
Init vmi, init_events: 1 init_paging 1 domain (null) domid 59 json (null)
Setup trace
Setup trace finished
VM Fork is ready for fuzzing
Starting fuzzer on 59
Running in standalone mode
Get 16 bytes of input from output/default/crashes/id:000000,sig:06,src:000000,time:2090,op:havoc,rep:8
Got input size 16
Starting trace from 0xffffffff81e07f58.
Writing 16 bytes of input to 0xffff8880b7c13000
Starting fuzz loop
[TRACER cpuid] RIP: 0xffffffff81e08069
CUID leaf 13371501
[TRACER cpuid] RIP: 0xffffffff81e08072
CUID leaf ffff8880
[TRACER cpuid] RIP: 0xffffffff81e082c8
CUID leaf 13371502
[TRACER cpuid] RIP: 0xffffffff81e082d0
CUID leaf ffff8880
[TRACER cpuid] RIP: 0xffffffff81e08306
CUID leaf 13371503
[TRACER cpuid] RIP: 0xffffffff81e0830f
CUID leaf ffff8880
[TRACER int3] RIP: 0xffffffff8149e050
Sink kasan_report! Tracer counter: 0. Crash: 1.
Stopping fuzz loop.
Result: crash
Closing tracer
##### DONE #####
root@ccg-cr fw-bld11:/home/cros/demo# stepper --domid 60 --limit 100000 --stop-on-address 0xffffffff8149e050 > stepper.log
root@ccg-cr fw-bld11:/home/cros/demo# cat stepper.log | less

```


Init vmi, init_events: 1 init_paging 1 domain (null) domid 60 json (null)

0: 0xffffffff81e07f58	mov rbx, rsi	00000000	48 89 f3 48 8d 7e 0c e8 fc 70 69 ff 0f b7 43	H..H.~...pi...C
1: 0xffffffff81e07f5b	lea rdi, [rsi + 0xc]	00000000	48 8d 7e 0c e8 fc 70 69 ff 0f b7 43 0c 89 85 60 ff ff ff	H.~...pi...C...
2: 0xffffffff81e07f5f	call 0xbc563106	00000000	e8 fc 70 69 ff 0f b7 43 0c 89 85 60 ff ff ff	..pi...C...`...
3: 0xffffffff8149f060	movabs rax, 0xffffffffffffffff	00000000	48 b8 ff ff ff ff ff 7f ff ff 55 48 89 e5 48	H.....UH..H
4: 0xffffffff8149f06a	push rbp	00000000	55 48 89 e5 48 8b 4d 08 48 39 c7 76 2e 48 8d	UH..H.M.H9.v.H.
5: 0xffffffff8149f06b	mov rbp, rsp	00000000	48 89 e5 48 8b 4d 08 48 39 c7 76 2e 48 8d 47	H..H.M.H9.v.H.G
6: 0xffffffff8149f06e	mov rcx, qword ptr [rbp + 8]	00000000	48 8b 4d 08 48 39 c7 76 2e 48 8d 47 01 48 89	H.M.H9.v.H.G.H.
7: 0xffffffff8149f072	cmp rdi, rax	00000000	48 39 c7 76 2e 48 8d 47 01 48 89 c6 83 e6 07	H9.v.H.G.H.....
8: 0xffffffff8149f075	jbe 0xbcecc035	00000000	76 2e 48 8d 47 01 48 89 c6 83 e6 07 74 30 48	v.H.G.H.....t0H
9: 0xffffffff8149f077	lea rax, [rdi + 1]	00000000	48 8d 47 01 48 89 c6 83 e6 07 74 30 48 ba 00	H.G.H.....t0H..
10: 0xffffffff8149f07b	mov rsi, rax	00000000	48 89 c6 83 e6 07 74 30 48 ba 00 00 00 00 00	H.....t0H.....
11: 0xffffffff8149f07e	and esi, 7	00000000	83 e6 07 74 30 48 ba 00 00 00 00 fc ff df	...t0H.....
12: 0xffffffff8149f081	je 0xbcecc037	00000000	74 30 48 ba 00 00 00 00 00 fc ff df 48 c1 e8	t0H.....H..
13: 0xffffffff8149f083	movabs rdx, 0xdffffc0000000000	00000000	48 ba 00 00 00 00 00 fc ff df 48 c1 e8 03 0f	H.....H....
14: 0xffffffff8149f08d	shr rax, 3	00000000	48 c1 e8 03 0f b6 04 10 84 c0 75 02 5d c3 40	H.....u.].@
15: 0xffffffff8149f091	movzx eax, byte ptr [rax + rdx]	00000000	0f b6 04 10 84 c0 75 02 5d c3 40 38 f0 0f 9e	u.].@8...
16: 0xffffffff8149f095	test al, al	00000000	84 c0 75 02 5d c3 40 38 f0 0f 9e c0 84 c0 74	..u.].@8.....t
17: 0xffffffff8149f097	jne 0xbcecc009	00000000	75 02 5d c3 40 38 f0 0f 9e c0 84 c0 74 f4 31	u.].@8.....t.1
18: 0xffffffff8149f099	pop rbp	00000000	5d c3 40 38 f0 0f 9e c0 84 c0 74 f4 31 d2 be	].@8.....t.1..
19: 0xffffffff8149f09a	ret	00000000	c3 40 38 f0 0f 9e c0 84 c0 74 f4 31 d2 be 02	..@8.....t.1...
20: 0xffffffff81e07f64	movzx eax, word ptr [rbx + 0xc]	00000000	0f b7 43 0c 89 85 60 ff ff ff 41 89 c4 89 85	..C...`...A....
21: 0xffffffff81e07f68	mov dword ptr [rbp - 0xa0], eax	00000000	89 85 60 ff ff ff 41 89 c4 89 85 30 ff ff ff	..`...A....0...
22: 0xffffffff81e07f6e	mov r12d, eax	00000000	41 89 c4 89 85 30 ff ff ff 85 c0 0f 84 00 11	A....0.....
23: 0xffffffff81e07f71	mov dword ptr [rbp - 0xd0], eax	00000000	89 85 30 ff ff ff 85 c0 0f 84 00 11 00 00 49	..0.....I
24: 0xffffffff81e07f77	test eax, eax	00000000	85 c0 0f 84 00 11 00 00 49 8d 46 18 48 89 c7	I.F.H..
25: 0xffffffff81e07f79	je 0xbcecd10b	00000000	0f 84 00 11 00 00 49 8d 46 18 48 89 c7 48 89	I.F.H..H.
26: 0xffffffff81e07f7f	lea rax, [r14 + 0x18]	00000000	49 8d 46 18 48 89 c7 48 89 85 48 ff ff e8	I.F.H..H..H....
27: 0xffffffff81e07f83	mov rdi, rax	00000000	48 89 c7 48 89 85 48 ff ff ff e8 ce 72 69 ff	H..H..H.....ri.
28: 0xffffffff81e07f86	mov qword ptr [rbp - 0xb8], rax	00000000	48 89 85 48 ff ff ff e8 ce 72 69 ff 48 8b 7d	H..H.....ri.H.}
29: 0xffffffff81e07f8d	call 0xbc5632d8	00000000	e8 ce 72 69 ff 48 8b 7d 90 49 8b 5e 18 e8 c1	..ri.H.}.I.^...
30: 0xffffffff8149f260	movabs rax, 0xffffffffffffffff	00000000	48 b8 ff ff ff ff ff 7f ff ff 55 48 89 e5 48	H.....UH..H
31: 0xffffffff8149f26a	push rbp	00000000	55 48 89 e5 48 8b 4d 08 48 39 c7 76 31 48 8d	UH..H.M.H9.v1H.
32: 0xffffffff8149f26b	mov rbp, rsp	00000000	48 89 e5 48 8b 4d 08 48 39 c7 76 31 48 8d 47	H..H.M.H9.v1H.G
33: 0xffffffff8149f26e	mov rcx, qword ptr [rbp + 8]	00000000	48 8b 4d 08 48 39 c7 76 31 48 8d 47 07 48 89	H.M.H9.v1H.G.H.
34: 0xffffffff8149f272	cmp rdi, rax	00000000	48 39 c7 76 31 48 8d 47 07 48 89 c2 83 e2 07	H9.v1H.G.H.....
35: 0xffffffff8149f275	jbe 0xbcecc038	00000000	76 31 48 8d 47 07 48 89 c2 83 e2 07 48 83 fa	v1H.G.H.....H..
36: 0xffffffff8149f277	lea rax, [rdi + 7]	00000000	48 8d 47 07 48 89 c2 83 e2 07 48 83 fa 07 75	H.G.H.....H...u
37: 0xffffffff8149f27b	mov rdx, rax	00000000	48 89 c2 83 e2 07 48 83 fa 07 75 2f 48 ba 00	H....H...u/H..
38: 0xffffffff8149f27e	and edx, 7	00000000	83 e2 07 48 83 fa 07 75 2f 48 ba 00 00 00 00	...H...u/H.....
39: 0xffffffff8149f281	cmp rdx, 7	00000000	48 83 fa 07 75 2f 48 ba 00 00 00 00 fc ff	H...u/H.....
40: 0xffffffff8149f285	jne 0xbcecc036	00000000	75 2f 48 ba 00 00 00 00 00 fc ff df 48 c1 e8	u/H.....H..

```
root@ccg-cr fw-bld11:/home/cros/demo# cat stepper.log | awk '{ print $2 }' | addr2line -e kfx-2.vmlinux -f -p -a > stepper-decoded.log
root@ccg-cr fw-bld11:/home/cros/demo#
```


GNU nano 4.8stepper-decoded.log

0xffffffff8149f277: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
0xffffffff8149f27b: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
0xffffffff8149f27e: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
0xffffffff8149f281: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
0xffffffff8149f285: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
0xffffffff8149f287: memory_is_poisoned_1 at /home/cros/chrome-kernel/mm/kasan/generic.c:51
0xffffffff8149f291: kasan_mem_to_shadow at /home/cros/chrome-kernel/./include/linux/kasan.h:28
0xffffffff8149f295: memory_is_poisoned_1 at /home/cros/chrome-kernel/mm/kasan/generic.c:51
0xffffffff8149f299: memory_is_poisoned_1 at /home/cros/chrome-kernel/mm/kasan/generic.c:53
0xffffffff8149f29b: memory_is_poisoned_1 at /home/cros/chrome-kernel/mm/kasan/generic.c:53
0xffffffff8149f29d: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
0xffffffff8149f29e: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
0xffffffff82212202: gro_pull_from_frag0 at /home/cros/chrome-kernel/net/core/dev.c:5422 (discriminator 2)
0xffffffff82212205: memcpy at /home/cros/chrome-kernel/./include/linux/string.h:403 (discriminator 2)
0xffffffff82212209: skb_tail_pointer at /home/cros/chrome-kernel/./include/linux/skbuff.h:2167 (discriminator 2)
0xffffffff8221220c: memcpy at /home/cros/chrome-kernel/./include/linux/string.h:403 (discriminator 2)
0xffffffff8221220f: skb_tail_pointer at /home/cros/chrome-kernel/./include/linux/skbuff.h:2167 (discriminator 2)
0xffffffff82212213: memcpy at /home/cros/chrome-kernel/./include/linux/string.h:403 (discriminator 2)
0xffffffff8149d740: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:121
0xffffffff8149d741: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:121
0xffffffff8149d744: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:121
0xffffffff8149d746: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:121
0xffffffff8149d749: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:121
0xffffffff8149d74b: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:122
0xffffffff8149d74f: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:121
0xffffffff8149d752: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:122
0xffffffff8149d755: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:121
0xffffffff8149d757: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:121
0xffffffff8149d75a: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:122
0xffffffff8149d75c: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:122
0xffffffff8149d75f: memcpy at /home/cros/chrome-kernel/mm/kasan/common.c:122
0xffffffff8149f530: check_memory_region_inline at /home/cros/chrome-kernel/mm/kasan/generic.c:172
0xffffffff8149f533: check_memory_region_inline at /home/cros/chrome-kernel/mm/kasan/generic.c:172
0xffffffff8149f539: check_memory_region_inline at /home/cros/chrome-kernel/mm/kasan/generic.c:175
0xffffffff8149f543: check_memory_region at /home/cros/chrome-kernel/mm/kasan/generic.c:190
0xffffffff8149f544: check_memory_region at /home/cros/chrome-kernel/mm/kasan/generic.c:190
0xffffffff8149f547: check_memory_region at /home/cros/chrome-kernel/mm/kasan/generic.c:190
0xffffffff8149f54a: check_memory_region at /home/cros/chrome-kernel/mm/kasan/generic.c:190
0xffffffff8149f54c: check_memory_region at /home/cros/chrome-kernel/mm/kasan/generic.c:190

^G Get Help
^X Exit

^O Write Out
^R Read File

^W Where Is
^_ Replace

^K Cut Text
^U Paste Text

^J Justify
^T To Spell

^C Cur Pos
^_ Go To Line

M-U Undo
M-E Redo

M-A Mark Text
M-6 Copy Text

M-] To Bracket
^Q Where Was

M-Q Previous
M-W Next


```

root@ccg-crfw-bld11:/home/cros/demo# cat stepper.log | awk '{ print $2 }' | addr2line -e kfx-2.vmlinux -f -p -a > stepper-decoded.log
root@ccg-crfw-bld11:/home/cros/demo# nano stepper-decoded.log
root@ccg-crfw-bld11:/home/cros/demo# xl list
Name                               ID    Mem VCPUs    State    Time(s)
Domain-0                           0 24000    12    r----- 5813.5
ubuntu-2004                         37  4030     1    --p---   24.7
(null)                             38     0     1    --p---    0.0
(null)                             39     0     1    r----- 464.1
(null)                             60     0     1    --p---    0.1
root@ccg-crfw-bld11:/home/cros/demo# forkvm 37
Fork VM id: 61
root@ccg-crfw-bld11:/home/cros/demo# rwmem --domid 61 --write 0xffff8880b7c13000 --file output/default/crashes/id\:000000\,sig\:06\,src\:000000\,time\:2090\,op\:havoc\,rep\
8 --limit 16
Init vmi, init_events: 0 init_paging 1 domain (null) domid 61 json (null)
Write operation success: 16 bytes to 0xffff8880b7c13000
root@ccg-crfw-bld11:/home/cros/demo# stepper --domid 61 --limit 100000 --stop-on-address 0xffffffff8149d75f > /dev/null
root@ccg-crfw-bld11:/home/cros/demo# xl list
Name                               ID    Mem VCPUs    State    Time(s)
Domain-0                           0 24000    12    r----- 5833.4
ubuntu-2004                         37  4030     1    --p---   24.7
(null)                             38     0     1    --p---    0.0
(null)                             39     0     1    --p---  472.7
(null)                             60     0     1    --p---    0.1
(null)                             61     0     1    --p---    0.1
root@ccg-crfw-bld11:/home/cros/demo# xen-hvmctx 61 | grep rsi
Header: magic 0x54381286, version 1
rbp 0xffff8880c9409bf0 rsi 0x000000000000001c
root@ccg-crfw-bld11:/home/cros/demo# xen-hvmctx 61 | grep rdi
rdi 0x0000000000000000 rsp 0xffff8880c9409bd8
root@ccg-crfw-bld11:/home/cros/demo#

```

Both *src* and *dst* in the memcpy is corrupted

Anything else we can do?

```

ubuntu-2004      37  4030    1  --p---    24.7
(null)           38    0    1  --p---     0.0
(null)           39    0    1  r-----   686.3
(null)           60    0    1  --p---     0.1
(null)           61    0    1  --p---     0.1
(null)           62    0    1  --p---     0.0
(null)           63    0    1  --p---     0.0
(null)           66    0    1  --p---     0.1
root@ccg-crfw-bld11:/home/cros/demo# cat setup.log
Init vmi, init_events: 1 init_paging 0 domain (null) domid 37 json (null)
Waiting for harness start (cpuid with leaf 0x13371500)
Got start cpuid callback with leaf: 0x13371500 subleaf: 0x10
Target buffer & size: 0xffff8880b7c13000 16
Parent is ready
root@ccg-crfw-bld11:/home/cros/demo# kfx --domid 37 --json kfx-2.json --address 0xffff8880b7c13000 --input input/seed --input-limit 16 --nocov --debug
##### START #####
Init vmi, init_events: 0 init_paging 0 domain (null) domid 37 json (null)
Parent is ready
Fork VMs created: 37 -> 67 -> 68
Init vmi, init_events: 0 init_paging 1 domain (null) domid 67 json kfx-2.json
Creating sink list from built-in information listed in sink.h
Setting breakpoint on sink ubsan_prologue 0xffffffff82502bcb -> 0x2502bcb
Setting breakpoint on sink kasan_report 0xffffffff8149e050 -> 0x149e050
Setting breakpoint on sink page_fault 0xffffffff82601250 -> 0x2601250
Setting breakpoint on sink oops_begin 0xffffffff81060560 -> 0x1060560
Setting breakpoint on sink panic 0xffffffff824d2049 -> 0x24d2049
Sinks are ready
Init vmi, init_events: 1 init_paging 1 domain (null) domid 68 json (null)
Setup trace
Setup trace finished
VM Fork is ready for fuzzing
Starting fuzzer on 68
Running in standalone mode
Get 16 bytes of input from input/seed
Got input size 16
Starting trace from 0xffffffff81e07f58.
Writing 16 bytes of input to 0xffff8880b7c13000
Starting fuzz loop
[TRACER cpuid] RIP: 0xffffffff81e08069
CPUID leaf 13371501
[TRACER cpuid] RIP: 0xffffffff81e08072
CPUID leaf ffff8880

```



```

Init vmi, init_events: 0 init_paging 1 domain (null) domid 67 json kfx-2.json
Creating sink list from built-in information listed in sink.h
Setting breakpoint on sink ubsan_prologue 0xffffffff82502bcb -> 0x2502bcb
Setting breakpoint on sink kasan_report 0xffffffff8149e050 -> 0x149e050
Setting breakpoint on sink page_fault 0xffffffff82601250 -> 0x2601250
Setting breakpoint on sink oops_begin 0xffffffff81060560 -> 0x1060560
Setting breakpoint on sink panic 0xffffffff824d2049 -> 0x24d2049
Sinks are ready
Init vmi, init_events: 1 init_paging 1 domain (null) domid 68 json (null)
Setup trace
Setup trace finished
VM Fork is ready for fuzzing
Starting fuzzer on 68
Running in standalone mode
Get 16 bytes of input from input/seed
Got input size 16
Starting trace from 0xffffffff81e07f58.
Writing 16 bytes of input to 0xffff8880b7c13000
Starting fuzz loop
[TRACER cpuid] RIP: 0xffffffff81e08069
CPUID leaf 13371501
[TRACER cpuid] RIP: 0xffffffff81e08072
CPUID leaf ffff8880
[TRACER cpuid] RIP: 0xffffffff81e082c8
CPUID leaf 13371502
[TRACER cpuid] RIP: 0xffffffff81e082d0
CPUID leaf ffff8880
[TRACER cpuid] RIP: 0xffffffff81e08306
CPUID leaf 13371503
[TRACER cpuid] RIP: 0xffffffff81e0830f
CPUID leaf ffff8880
[TRACER cpuid] RIP: 0xffffffff81e07f03
CPUID leaf 13371337
    Harness signal on finish
Stopping fuzz loop.
Result: no crash
Closing tracer
##### DONE #####
root@ccg-crfw-bld11:/home/cros/demo# forkvm 37
Fork VM id: 69
root@ccg-crfw-bld11:/home/cros/demo# stepper --domid 69 --limit 1000000 --stop-on-address 0xffffffff81e07f03 > normal.log
root@ccg-crfw-bld11:/home/cros/demo# cat normal.log | awk '{ print $2 }' | addr2line -e kfx-2.vmlinux -f -p -a > normal-decoded.log
root@ccg-crfw-bld11:/home/cros/demo# diff stepper-decoded.log normal-decoded.log | less

```


19742,20206d19741

```
< 0xffffffff81e092c6: igb_build_skb at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_main.c:8104
< 0xffffffff81e092c9: igb_build_skb at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_main.c:8104
< 0xffffffff8149f260: check_memory_region_inline at /home/cros/chrome-kernel/mm/kasan/generic.c:175
< 0xffffffff8149f26a: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
< 0xffffffff8149f26b: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
< 0xffffffff8149f26e: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
< 0xffffffff8149f272: check_memory_region_inline at /home/cros/chrome-kernel/mm/kasan/generic.c:175
< 0xffffffff8149f275: check_memory_region_inline at /home/cros/chrome-kernel/mm/kasan/generic.c:175
< 0xffffffff8149f277: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
< 0xffffffff8149f27b: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
< 0xffffffff8149f27e: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
< 0xffffffff8149f281: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
< 0xffffffff8149f285: memory_is_poisoned_2_4_8 at /home/cros/chrome-kernel/mm/kasan/generic.c:70
< 0xffffffff8149f287: memory_is_poisoned_1 at /home/cros/chrome-kernel/mm/kasan/generic.c:51
< 0xffffffff8149f291: kasan_mem_to_shadow at /home/cros/chrome-kernel/./include/linux/kasan.h:28
< 0xffffffff8149f295: memory_is_poisoned_1 at /home/cros/chrome-kernel/mm/kasan/generic.c:51
< 0xffffffff8149f299: memory_is_poisoned_1 at /home/cros/chrome-kernel/mm/kasan/generic.c:53
< 0xffffffff8149f29b: memory_is_poisoned_1 at /home/cros/chrome-kernel/mm/kasan/generic.c:53
< 0xffffffff8149f29d: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
< 0xffffffff8149f29e: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
< 0xffffffff81e092ce: igb_build_skb at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_main.c:8104
< 0xffffffff81e092d5: igb_build_skb at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_main.c:8104
< 0xffffffff81e092d8: igb_build_skb at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_main.c:8104
< 0xffffffff81e092db: igb_build_skb at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_main.c:8104
< 0xffffffff81e29f60: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f65: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f66: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f69: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f6b: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f6d: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f6f: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f72: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f74: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f77: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f78: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f7b: skb_end_pointer at /home/cros/chrome-kernel/./include/linux/skbuff.h:1385
< 0xffffffff81e29f82: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:871
< 0xffffffff81e29f86: igb_ptp_rx_pktstamp at /home/cros/chrome-kernel/drivers/net/ethernet/intel/igb/igb_ptp.c:873
< 0xffffffff8149f260: check_memory_region_inline at /home/cros/chrome-kernel/mm/kasan/generic.c:175
< 0xffffffff8149f26a: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
< 0xffffffff8149f26b: __asan_load8 at /home/cros/chrome-kernel/mm/kasan/generic.c:251
```

:

Buggy code:

```
/* pull timestamp out of packet data */  
if (igb_test_staterr(rx_desc, E1000_RXDADV_STAT_TSIP)) {  
    igb_ptp_rx_pktstamp(rx_ring->q_vector, skb->data, skb);  
    __skb_pull(skb, IGB_TS_HDR_LEN);  
}
```

Fixed code:

```
/* pull timestamp out of packet data */  
if (igb_test_staterr(rx_desc, E1000_RXDADV_STAT_TSIP)) {  
    if (!igb_ptp_rx_pktstamp(rx_ring->q_vector, skb->data, skb))  
        __skb_pull(skb, IGB_TS_HDR_LEN);  
}
```

can you spot the bug?

```
case TRB_RESET_DEV:
    /* SLOT_ID field in reset device cmd completion event TRB is 0.
     * Use the SLOT_ID from the command TRB instead (xhci 4.6.11)
     */
    slot_id = TRB_TO_SLOT_ID(
        le32_to_cpu(cmd_trb->generic.field[3]));
    xhci_handle_cmd_reset_dev(xhci, slot_id, event);
    break;
```

```
static void xhci_handle_cmd_reset_dev(struct xhci_hcd *xhci, int slot_id,
                                     struct xhci_event_cmd *event)
{
    struct xhci_virt_device *vdev;
    struct xhci_slot_ctx *slot_ctx;

    vdev = xhci->devs[slot_id];
    slot_ctx = xhci_get_slot_ctx(xhci, vdev->out_ctx);
    trace_xhci_handle_cmd_reset_dev(slot_ctx);

    xhci_dbg(xhci, "Completed reset device command.\n");
    if (!xhci->devs[slot_id])
        xhci_warn(xhci, "Reset device command completion "
                  "for disabled slot %u\n", slot_id);
}
```


what can go wrong here?

```
static void handle_cmd_completion(struct xhci_hcd *xhci,  
                                struct xhci_event_cmd *event)  
{  
    int slot_id = TRB_TO_SLOT_ID(le32_to_cpu(event->flags));
```

```
    cmd_type = TRB_FIELD_TO_TYPE(le32_to_cpu(cmd_trb->generic.field[3]));  
    switch (cmd_type) {
```

```
    case TRB_ADDR_DEV:  
        xhci_handle_cmd_addr_dev(xhci, slot_id);  
        break;
```

```
static void xhci_handle_cmd_addr_dev(struct xhci_hcd *xhci, int slot_id)  
{  
    struct xhci_virt_device *vdev;  
    struct xhci_slot_ctx *slot_ctx;  
  
    vdev = xhci->devs[slot_id];  
    slot_ctx = xhci_get_slot_ctx(xhci, vdev->out_ctx);  
    trace_xhci_handle_cmd_addr_dev(slot_ctx);  
}
```

what we found & fixed

- 9 NULL-pointer derefs
- 3 array index out-of-bounds
- 2 infinite loops in IRQ handler
- 2 user-memory accesses

```
BUG: KASAN: null-ptr-deref in handle_cmd_completion+0xde7/0x2030
Read of size 8 at addr 0000000000000008 by task kworker/0:0/5
1171 ep_ctx = xhci_get_ep_ctx(xhci, vdev->out_ctx, ep_index);
```

```
BUG: KASAN: null-ptr-deref in xhci_stream_id_to_ring+0x3b/0xc0
Read of size 8 at addr 00000000000000c8 by task kworker/0:0/5
595 if (!ep->stream_info)
```

```
2029 while (ep_ring->dequeue != td->last_trb)
2030     inc_deq(xhci, ep_ring);
```

```
BUG: KASAN: user-memory-access in xhci_stream_id_to_ring+0x8b/0xc0
Read of size 8 at addr 0000000000001080 by task kworker/0:0/5
594 return ep->ring;
```

```
BUG: KASAN: user-memory-access in xhci_stream_id_to_ring+0x3b/0xc0
Read of size 8 at addr 0000000000001328 by task kworker/0:0/5
595 if (!ep->stream_info)
```

```
UBSAN: Undefined behaviour in drivers/usb/host/xhci-ring.c:1199:47
index 4294967295 is out of range for type 'xhci_virt_ep [31]'
1199 xhci->devs[slot_id]->eps[ep_index].ep_state &= ~EP_HALTED;
```

```
UBSAN: Undefined behaviour in drivers/usb/host/xhci-ring.c:1081:16
index 4294967295 is out of range for type 'xhci_virt_ep [31]'
1081 ep = &dev->eps[ep_index];
```

doublefetch detection

Remove R/W EPT permission of DMA page

- Byte-granularity
- Look for PF with read access at the same page & offset 2x
- Report crash to AFL

We thought it would be rare

- Happens all over the place
- Happens because of different bits used of the same byte
- Doesn't make sense to stop execution at the first detection

mission accomplished!!

Or is it?

- Did we discover all DMA input points?
- We got bogged down by documenting all the bugs
- Staring at the code is only productive up to a point

Let's do better!

dmamonitor

Standalone EPT fault monitoring

Linux allocates DMA pages through its internal DMA API

- `dma_alloc_attrs()`
- Hook function entry & ret with `int3`

Remove EPT permission on-the-fly

- Log all code-sites that read from any DMA!

GNU GRUB version 2.04

*Ubuntu

Advanced options for Ubuntu

Memory test (memtest86+)

Memory test (memtest86+, serial console 115200)

Zephyr Kernel

Use the ^ and v keys to select which entry is highlighted.

Press enter to boot the selected OS, 'e' to edit the commands
before booting or 'c' for a command-line.

The highlighted entry will be executed automatically in 1s.

root@t2:/shared/xhci_demo# screen -L dmamonitor --domain ubuntu-20.04 --json kernel.json


```

1.411137] ACPI: bus type USB registered
1.412193] usbcore: registered new interface driver usbfs
1.416141] usbcore: registered new interface driver hub
1.420126] usbcore: registered new device driver usb
1.424257] pps_core: LinuxPPS API ver. 1 registered
1.428081] pps_core: Software ver. 5.3.6 - Copyright 2005-2007 Rodolfo Giometti <giometti@linux.it>
1.432120] PTP clock support registered
1.435432] EDAC MC: Ver: 3.0.0
1.438044] NetLabel: Initializing
1.440082] NetLabel: domain hash size = 128
1.443319] NetLabel: protocols = UNLABELED CIPSOv4 CALIPSO
1.444199] NetLabel: unlabeled traffic allowed by default
1.448243] PCI: Using ACPI for IRQ routing
1.452206] hpet: 3 channels of 0 reserved for per-cpu timers
1.456095] hpet0: at MMIO 0xfed00000, IRQs 2, 8, 0
1.459735] hpet0: 3 comparators, 64-bit 62.500000 MHz counter
1.462127] clocksource: Switched to clocksource xen
1.464558] VFS: Disk quotas dquot_6.6.0
1.467495] VFS: Dquot-cache hash table entries: 512 (order 0, 4096 bytes)
1.473402] AppArmor: AppArmor Filesystem Enabled
1.476885] pnp: PnP ACPI init
1.479720] system 00:00: [mem 0x00000000-0x0009ffff] could not be reserved
1.485831] system 00:01: [io 0x08a0-0x08a3] has been reserved
1.490796] system 00:01: [io 0x0cc0-0x0ccf] has been reserved
1.495168] system 00:01: [io 0x04d0-0x04d1] has been reserved
1.502256] system 00:07: [io 0xae00-0xae0f] has been reserved
1.506634] system 00:07: [io 0xb044-0xb047] has been reserved
1.518844] pnp: PnP ACPI: found 8 devices
1.535039] clocksource: acpi_pm: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 2085701024 ns
1.542084] NET: Registered protocol family 2
1.546153] tcp_listen_portaddr_hash hash table entries: 512 (order: 1, 8192 bytes, linear)
1.552696] TCP established hash table entries: 8192 (order: 4, 65536 bytes, linear)
1.558795] TCP bind hash table entries: 8192 (order: 5, 131072 bytes, linear)
1.564257] TCP: Hash tables configured (established 8192 bind 8192)
1.569550] MPTCP token hash table entries: 1024 (order: 2, 24576 bytes, linear)
1.575071] UDP hash table entries: 512 (order: 2, 16384 bytes, linear)
1.580260] UDP-Lite hash table entries: 512 (order: 2, 16384 bytes, linear)
1.585911] NET: Registered protocol family 1
1.589407] NET: Registered protocol family 44
1.592849] pci_bus 0000:00: resource 4 [io 0x0000-0x0cf7 window]
1.597608] pci_bus 0000:00: resource 5 [io 0x0d00-0xffff window]
1.602679] pci_bus 0000:00: resource 6 [mem 0x000a0000-0x000bffff window]
1.608039] pci_bus 0000:00: resource 7 [mem 0xf0000000-0xfbffffff window]
1.613680] pci 0000:00:01.0: PIIX3: Enabling Passive Release
1.617932] pci 0000:00:00.0: Limiting direct PCI/PCI transfers
1.622481] pci 0000:00:01.0: Activating ISA DMA hang workarounds
1.634853] PCI: CLS 0 bytes, default 64
1.637854] Trying to unpack rootfs image as initramfs...

```

```

Init vmi, init_events: 1 init_paging 0 domain ubuntu-20.04 domid 0 json (null)
dma_alloc_attrs @ 0xffffffff811daa90

```

I


```

2.814905] uhci_hcd: USB Universal Host Controller Interface driver
2.821379] xen_netfront: backend supports XDP headroom
2.838849] xhci_hcd 0000:00:04.0: xHCI Host Controller
2.842923] xhci_hcd 0000:00:04.0: new USB bus registered, assigned bus number 1
3.456952] xhci_hcd 0000:00:04.0: hcc params 0x00087001 hci version 0x100 quirks 0x0000000000000010
3.466305] usb usb1: New USB device found, idVendor=1d6b, idProduct=0002, bcdDevice= 5.10
3.472347] usb usb1: New USB device strings: Mfr=3, Product=2, SerialNumber=1
3.477850] usb usb1: Product: xHCI Host Controller
3.481660] usb usb1: Manufacturer: Linux 5.10.0-xhci-kfx-build1 xhci-hcd
3.486833] usb usb1: SerialNumber: 0000:00:04.0
3.490980] hub 1-0:1.0: USB hub found
3.493925] hub 1-0:1.0: 4 ports detected
3.499007] xhci_hcd 0000:00:04.0: xHCI Host Controller
3.502563] xhci_hcd 0000:00:04.0: new USB bus registered, assigned bus number 2
3.507880] xhci_hcd 0000:00:04.0: Host supports USB 3.0 SuperSpeed
3.512704] usb usb2: We don't know the algorithms for LPM for this host, disabling LPM.
3.518908] usb usb2: New USB device found, idVendor=1d6b, idProduct=0003, bcdDevice= 5.10
3.525104] usb usb2: New USB device strings: Mfr=3, Product=2, SerialNumber=1
3.530443] usb usb2: Product: xHCI Host Controller
3.534060] usb usb2: Manufacturer: Linux 5.10.0-xhci-kfx-build1 xhci-hcd
3.539359] usb usb2: SerialNumber: 0000:00:04.0
3.543582] hub 2-0:1.0: USB hub found
3.546798] hub 2-0:1.0: 8 ports detected
3.553166] usbcore: registered new interface driver usb-storage
3.557951] i8042: PNP: PS/2 Controller [PNP0303:PS2K,PNP0f13:PS2M] at 0x60,0x64 irq 1,12
3.566121] serio: i8042 KBD port at 0x60,0x64 irq 1
3.569959] serio: i8042 AUX port at 0x60,0x64 irq 12
3.574138] mousedev: PS/2 mouse device common for all mice
3.579843] input: AT Translated Set 2 keyboard as /devices/platform/i8042/serio0/input/input2
3.587231] rtc_cmos 00:02: registered as rtc0
3.590731] rtc_cmos 00:02: setting system clock to 2021-06-21T17:31:55 UTC (1624296715)
3.597578] rtc_cmos 00:02: alarms up to one day, 114 bytes nvram, hpet irqs
3.603535] i2c /dev entries driver
3.606441] device-mapper: uevent: version 1.0.3
3.610163] device-mapper: ioctl: 4.43.0-ioctl (2020-10-01) initialised: dm-devel@redhat.com
3.616151] platform eisa.0: Probing EISA bus 0
3.619359] platform eisa.0: EISA: Cannot allocate resource for mainboard
3.624280] platform eisa.0: Cannot allocate resource for EISA slot 1
3.628895] platform eisa.0: Cannot allocate resource for EISA slot 2
3.633618] platform eisa.0: Cannot allocate resource for EISA slot 3
3.638649] platform eisa.0: Cannot allocate resource for EISA slot 4
3.643650] platform eisa.0: Cannot allocate resource for EISA slot 5
3.648590] platform eisa.0: Cannot allocate resource for EISA slot 6
3.653159] platform eisa.0: Cannot allocate resource for EISA slot 7
3.658796] platform eisa.0: Cannot allocate resource for EISA slot 8
3.663387] platform eisa.0: EISA: Detected 0 cards
3.667120] intel_pstate: CPU model not supported
3.670760] ledtrig-cpu: registered to indicate activity on CPUs
3.675329] intel_pmc_core intel_pmc_core

```

```

DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affd5 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affd6 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affd7 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affd8 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affd9 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affda rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affdb rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affdc rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affdd rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affde rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affdf rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe0 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe1 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe2 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe3 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe4 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe5 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe6 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe7 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe8 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affe9 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affea rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affeb rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affec rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affed rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affee rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affef rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff0 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff1 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff2 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff3 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff4 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff5 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff6 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff7 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff8 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afff9 rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afffa rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afffb rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afffc rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afffd rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338afffe rw
DMA access! RIP: 0xfffffffff818f5599 Mem: 0xfffff8880338affff rw
DMA allocated @ 0xfffff8880338b0000. Size: 16. Dev: 0xfffff88800656c0b8
EPT permissions changed for page at 0x338b0
DMA access! RIP: 0xfffffffff81d7c162 Mem: 0xfffff8880338b0000 rw
DMA access! RIP: 0xfffffffff81d7c16e Mem: 0xfffff8880338b0008 rw
DMA access! RIP: 0xfffffffff81d7c17e Mem: 0xfffff8880338b000c rw

```



```

Mounting Mount unit for gnome-3-34-1804, revision 66...
Mounting Mount unit for gnome-3-34-1804, revision 72...
Mounting Mount unit for go, revision 7696...
Mounting Mount unit for go, revision 7736...
Mounting Mount unit for gtk-common-themes, revision 1514...
Mounting Mount unit for gtk-common-themes, revision 1515...
Mounting Mount unit for snap-store, revision 542...
Mounting Mount unit for snap-store, revision 547...
Mounting Mount unit for snapd, revision 12057...
Mounting Mount unit for snapd, revision 12159...
Starting udev Kernel Device Manager...
[ OK ] Finished udev Coldplug all Devices.
[ OK ] Mounted Mount unit for core18, revision 1997.
[ OK ] Mounted Mount unit for core18, revision 2066.
[ OK ] Mounted Mount unit for gnome-3-34-1804, revision 66.
[ OK ] Mounted Mount unit for gnome-3-34-1804, revision 72.
[ OK ] Mounted Mount unit for go, revision 7696.
[ OK ] Mounted Mount unit for go, revision 7736.
[ OK ] Mounted Mount unit for gtk-common-themes, revision 1514.
[ OK ] Mounted Mount unit for gtk-common-themes, revision 1515.
[ OK ] Mounted Mount unit for snap-store, revision 542.
[ OK ] Mounted Mount unit for snap-store, revision 547.
[ OK ] Mounted Mount unit for snapd, revision 12057.
[ OK ] Mounted Mount unit for snapd, revision 12159.
[ OK ] Started udev Kernel Device Manager.
[ OK ] Started Dispatch Password ...ts to Console Directory Watch.
[ OK ] Reached target Local Encrypted Volumes.
[ OK ] Found device /dev/ttyS0.
[ OK ] Found device /dev/disk/by-uuid/66A8-01DE.
Starting File System Check.../dev/disk/by-uuid/66A8-01DE...
[ OK ] Started File System Check Daemon to report status.
[ OK ] Finished File System Check...n /dev/disk/by-uuid/66A8-01DE.
Mounting /boot/efi...
[ OK ] Mounted /boot/efi.
[ OK ] Reached target Local File Systems.
[ OK ] Listening on Load/Save RF ...itch Status /dev/rfkill Watch.
Starting Load AppArmor profiles...
Starting Set console font and keymap...
Starting Tell Plymouth To Write Out Runtime Data...
Starting Create Volatile Files and Directories...
[ OK ] Finished Set console font and keymap.
[ OK ] Finished Tell Plymouth To Write Out Runtime Data.
[ OK ] Found device /dev/hvc0.
[ OK ] Finished Create Volatile Files and Directories.
Starting Network Name Resolution...
Starting Network Time Synchronization...
Starting Update UTMP about System Boot/Shutdown...
[ OK ] Finished Update UTMP about System Boot/Shutdown.

```

```

root@t2:/shared/xhci_demo# screen -L dmamonitor --domain ubuntu-20.04 --json kernel.json
[detached from 138232.pts-1.t2]
root@t2:/shared/xhci_demo# cat screenlog.0 | grep "DMA alloc"
DMA allocated @ 0xffff8880338ac000. Size: 2056. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff8880338ad000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff8880338ae000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff8880338af000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff8880338b0000. Size: 16. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff8880101e1000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff888005b1c000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff888005b1f000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff8880072b3000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff888011da4000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff88801412b000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff888010145000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff88800c8d2000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff8880100a6000. Size: 4096. Dev: 0xffff88800656c0b8
DMA allocated @ 0xffff88801057c000. Size: 4096. Dev: 0xffff88800656c0b8
root@t2:/shared/xhci_demo#

```

[OK]	Started System Logging Service.	0xfffffffff81d7ade4
[OK]	Started LSB: Record successful boot for GRUB.	0xfffffffff81d7ae00
	Starting GRUB failed boot detection...	0xfffffffff81d7ae1c
[OK]	Started LSB: automatic crash report generation.	0xfffffffff81d7af29
[OK]	Started WPA supplicant.	0xfffffffff81d7af42
[OK]	Started Dispatcher daemon for systemd-networkd.	0xfffffffff81d7af5e
[OK]	Finished GRUB failed boot detection.	0xfffffffff81d7af7a
[OK]	Started Avahi mDNS/DNS-SD Stack.	0xfffffffff81d7ce88
[OK]	Started Login Service.	0xfffffffff81d7eac5
[OK]	Started Network Manager.	0xfffffffff81d7eaff
[OK]	Started Authorization Manager.	0xfffffffff81d7f981
[OK]	Started Switcheroo Control Proxy service.	0xfffffffff81d7fcc1
[OK]	Reached target Network.	0xfffffffff81d8095c
	Starting Modem Manager...	0xfffffffff81d8098d
	Starting Network Manager Wait Online...	0xfffffffff81d82b8c
[OK]	Started Make remote CUPS printers available locally.	0xfffffffff81d837c7
	Starting OpenVPN service...	0xfffffffff81d85259
	Starting OpenBSD Secure Shell server...	0xfffffffff81d8527b
	Starting Hostname Service...	0xfffffffff81d8535c
	Starting Permit User Sessions...	0xfffffffff81d853d1
[OK]	Started Unattended Upgrades Shutdown.	0xfffffffff81d86f3b
[OK]	Finished OpenVPN service.	0xfffffffff81d86f76
[OK]	Finished Permit User Sessions.	0xfffffffff81d86fb2
[OK]	Started OpenBSD Secure Shell server.	0xfffffffff81d873a7
[OK]	Started Accounts Service.	0xfffffffff81d8743d
[OK]	Started Modem Manager.	0xfffffffff81d875e0
[OK]	Started Disk Manager.	0xfffffffff81d875fb
	Starting GNOME Display Manager...	0xfffffffff81d8789e
	Starting Hold until boot process finishes up...	0xfffffffff81d878a9
[OK]	Finished Hold until boot process finishes up.	0xfffffffff81d87973
[OK]	Started Hostname Service.	0xfffffffff81d87c77
	Starting Network Manager Script Dispatcher Service...	0xfffffffff81d87ce9
[OK]	Started Serial Getty on hvc0.	0xfffffffff81d87d39
[OK]	Started Serial Getty on ttyS0.	0xfffffffff81d87d4c
[OK]	Reached target Login Prompts.	0xfffffffff81d87d6b
	Starting Set console scheme...	0xfffffffff81d87da5
[OK]	Finished Network Manager Wait Online.	0xfffffffff81d88272
[OK]	Finished Set console scheme.	0xfffffffff81d88b04
[OK]	Started Network Manager Script Dispatcher Service.	0xfffffffff81d88b83
[OK]	Created slice system-getty.slice.	0xfffffffff81d9d773
[OK]	Reached target Network is Online.	0xfffffffff81d9d8af
	Starting Tool to automatic...mit kernel crash signatures...	0xfffffffff81d9d8e4
[OK]	Started crash report submission daemon.	0xfffffffff81d9d900
[OK]	Started Tool to automatic...ubmit kernel crash signatures.	0xfffffffff81d9d92b
[OK]	Reached target Multi-User System.	0xfffffffff81d9da48
		0xfffffffff81d9dacd
		0xfffffffff81d9de0c
Ubuntu 20.04.2 LTS ubuntu ttyS0		root@t2:/shared/xhci_demo# cat screenlog.0 grep "r-" awk '{ print \$4 }' sort -u addr2line -e ../
ubuntu login:		linux/vmlinux -f -p -a


```

[ OK ] Started System Logging Service.
[ OK ] Started LSB: Record successful boot for GRUB.
      Starting GRUB failed boot detection...
[ OK ] Started LSB: automatic crash report generation.
[ OK ] Started WPA supplicant.
[ OK ] Started Dispatcher daemon for systemd-networkd.
[ OK ] Finished GRUB failed boot detection.
[ OK ] Started Avahi mDNS/DNS-SD Stack.
[ OK ] Started Login Service.
[ OK ] Started Network Manager.
[ OK ] Started Authorization Manager.
[ OK ] Started Switcheroo Control Proxy service.
[ OK ] Reached target Network.
      Starting Modem Manager...
      Starting Network Manager Wait Online...
[ OK ] Started Make remote CUPS printers available locally.
      Starting OpenVPN service...
      Starting OpenBSD Secure Shell server...
      Starting Hostname Service...
      Starting Permit User Sessions...
[ OK ] Started Unattended Upgrades Shutdown.
[ OK ] Finished OpenVPN service.
[ OK ] Finished Permit User Sessions.
[ OK ] Started OpenBSD Secure Shell server.
[ OK ] Started Accounts Service.
[ OK ] Started Modem Manager.
[ OK ] Started Disk Manager.
      Starting GNOME Display Manager...
      Starting Hold until boot process finishes up...
[ OK ] Finished Hold until boot process finishes up.
[ OK ] Started Hostname Service.
      Starting Network Manager Script Dispatcher Service...
[ OK ] Started Serial Getty on hvc0.
[ OK ] Started Serial Getty on ttyS0.
[ OK ] Reached target Login Prompts.
      Starting Set console scheme...
[ OK ] Finished Network Manager Wait Online.
[ OK ] Finished Set console scheme.
[ OK ] Started Network Manager Script Dispatcher Service.
[ OK ] Created slice system-getty.slice.
[ OK ] Reached target Network is Online.
      Starting Tool to automatic...mit kernel crash signatures...
[ OK ] Started crash report submission daemon.
[ OK ] Started Tool to automatica...ubmit kernel crash signatures.
[ OK ] Reached target Multi-User System.

Ubuntu 20.04.2 LTS ubuntu ttyS0
ubuntu login: 

```

```

0xffffffff81d6cf84: xhci_add_endpoint at /shared/linux/drivers/usb/host/xhci.c:1904
0xffffffff81d6cf99: xhci_add_endpoint at /shared/linux/drivers/usb/host/xhci.c:1913
0xffffffff81d7290e: xhci_check_bandwidth at /shared/linux/drivers/usb/host/xhci.c:2900
0xffffffff81d72926: xhci_check_bandwidth at /shared/linux/drivers/usb/host/xhci.c:2901
0xffffffff81d729c0: xhci_check_bandwidth at /shared/linux/drivers/usb/host/xhci.c:2915
0xffffffff81d729d9: xhci_check_bandwidth at /shared/linux/drivers/usb/host/xhci.c:2916
0xffffffff81d72a33: xhci_check_bandwidth at /shared/linux/drivers/usb/host/xhci.c:2930
0xffffffff81d75c22: xhci_discover_or_reset_device at /shared/linux/drivers/usb/host/xhci.c:3750
0xffffffff81d7638d: xhci_setup_device at /shared/linux/drivers/usb/host/xhci.c:4128
0xffffffff81d7651c: xhci_setup_device at /shared/linux/drivers/usb/host/xhci.c:4209
0xffffffff81d765c1: xhci_setup_device at /shared/linux/drivers/usb/host/xhci.c:4225
0xffffffff81d765ed: xhci_setup_device at /shared/linux/drivers/usb/host/xhci.c:4227
0xffffffff81d7784d: xhci_link_segments at /shared/linux/drivers/usb/host/xhci-mem.c:114
0xffffffff81d79b93: xhci_setup_addressable_virt_dev at /shared/linux/drivers/usb/host/xhci-mem.c:1121
0xffffffff81d7adcb: xhci_endpoint_copy at /shared/linux/drivers/usb/host/xhci-mem.c:1627
0xffffffff81d7ade4: xhci_endpoint_copy at /shared/linux/drivers/usb/host/xhci-mem.c:1628
0xffffffff81d7ae00: xhci_endpoint_copy at /shared/linux/drivers/usb/host/xhci-mem.c:1629
0xffffffff81d7ae1c: xhci_endpoint_copy at /shared/linux/drivers/usb/host/xhci-mem.c:1630
0xffffffff81d7af29: xhci_slot_copy at /shared/linux/drivers/usb/host/xhci-mem.c:1652
0xffffffff81d7af42: xhci_slot_copy at /shared/linux/drivers/usb/host/xhci-mem.c:1653
0xffffffff81d7af5e: xhci_slot_copy at /shared/linux/drivers/usb/host/xhci-mem.c:1654
0xffffffff81d7af7a: xhci_slot_copy at /shared/linux/drivers/usb/host/xhci-mem.c:1655
0xffffffff81d7ce88: xhci_mem_init at /shared/linux/drivers/usb/host/xhci-mem.c:2437
0xffffffff81d7eac5: inc_enq at /shared/linux/drivers/usb/host/xhci-ring.c:210
0xffffffff81d7eaff: trb_is_link at /shared/linux/drivers/usb/host/xhci-ring.c:87
0xffffffff81d7f981: trb_is_link at /shared/linux/drivers/usb/host/xhci-ring.c:87
0xffffffff81d7fcc1: prepare_transfer at /shared/linux/drivers/usb/host/xhci-ring.c:3053
0xffffffff81d8095c: trb_is_link at /shared/linux/drivers/usb/host/xhci-ring.c:87
0xffffffff81d8098d: trb_is_link at /shared/linux/drivers/usb/host/xhci-ring.c:87
0xffffffff81d82b8c: giveback_first_trb at /shared/linux/drivers/usb/host/xhci-ring.c:3147
0xffffffff81d837c7: giveback_first_trb at /shared/linux/drivers/usb/host/xhci-ring.c:3147
0xffffffff81d85259: handle_cmd_completion at /shared/linux/drivers/usb/host/xhci-ring.c:1401
0xffffffff81d8527b: handle_cmd_completion at /shared/linux/drivers/usb/host/xhci-ring.c:1409
0xffffffff81d8535c: handle_cmd_completion at /shared/linux/drivers/usb/host/xhci-ring.c:1430
0xffffffff81d853d1: handle_cmd_completion at /shared/linux/drivers/usb/host/xhci-ring.c:1459
0xffffffff81d86f3b: finish_td at /shared/linux/drivers/usb/host/xhci-ring.c:1980
0xffffffff81d86f76: finish_td at /shared/linux/drivers/usb/host/xhci-ring.c:1983
0xffffffff81d86fb2: finish_td at /shared/linux/drivers/usb/host/xhci-ring.c:1985
0xffffffff81d873a7: xhci_handle_event at /shared/linux/drivers/usb/host/xhci-ring.c:2737
0xffffffff81d8743d: xhci_handle_event at /shared/linux/drivers/usb/host/xhci-ring.c:2752
0xffffffff81d875e0: handle_port_status at /shared/linux/drivers/usb/host/xhci-ring.c:1604
0xffffffff81d875fb: handle_port_status at /shared/linux/drivers/usb/host/xhci-ring.c:1609
0xffffffff81d8789e: handle_tx_event at /shared/linux/drivers/usb/host/xhci-ring.c:2359
0xffffffff81d878a9: handle_tx_event at /shared/linux/drivers/usb/host/xhci-ring.c:2359
0xffffffff81d87973: handle_tx_event at /shared/linux/drivers/usb/host/xhci-ring.c:2372
0xffffffff81d87c77: trb_is_noop at /shared/linux/drivers/usb/host/xhci-ring.c:82
0xffffffff81d87ce9: process_bulk_intr_td at /shared/linux/drivers/usb/host/xhci-ring.c:2269
0xffffffff81d87d39: process_bulk_intr_td at /shared/linux/drivers/usb/host/xhci-ring.c:2271
0xffffffff81d87d4c: process_bulk_intr_td at /shared/linux/drivers/usb/host/xhci-ring.c:2272

```

still some problems

Sometimes the data isn't used where DMA is accessed

- Where is it used and is it safe?

¯_(\ツ)_/¯

We don't want to go back to reading the source

- Very hard to follow data-flow by eye
- Error-prone, manual, annoying

full-vm taint analysis

Track tainted data propagation in the kernel

- Record kernel execution using VMtrace
- Replay recorded instruction stream in Triton DBI's taint engine
- Check where RIP gets tainted

Can find code-locations affected by DMA input anywhere


```

root@t2:/shared/kernel-fuzzer-for-xen-project# xl list
Name                ID    Mem VCPUs    State    Time(s)
Domain-0            0   8192     8    r----- 96503.0
ubuntu-20.04        86     0     1    --p---   11.9
root@t2:/shared/kernel-fuzzer-for-xen-project# forkvm 86
Fork VM id: 91
root@t2:/shared/kernel-fuzzer-for-xen-project# ./dmamonitor --domid 91 --json /shared/cfg/5.4.
0-48.json --dma 0xffffffffc02e0010
Init vmi, init_events: 1 init_paging 0 domain (null) domid 91 json (null)
dma_alloc_attrs @ 0xffffffff8112df40
EPT permissions changed for page at 0x14127a
DMA access! RIP: 0xffffffffc02de0c5 Mem: 0xffffffffc02e0010 r-

```

I

```

root@t2:/shared/vmtaint# xl list
Name                ID    Mem VCPUs    State    Time(s)
Domain-0            0   8192     8    r----- 96524.2
ubuntu-20.04        86     0     1    --p---   11.9
(null)              91     0     1    --p---    0.0
root@t2:/shared/vmtaint# xl unpause 91
libxl: error: libxl_xshelp.c:201:libxl__xs_read_mandatory: xenstore read failed: `/libxl/91/type': No suc
h file or directory
libxl: warning: libxl_dom.c:51:libxl__domain_type: unable to get domain type for domid=91, assuming HVM
root@t2:/shared/vmtaint# 

```

```

root@t2:/shared/kernel-fuzzer-for-xen-project# forkvm 86
Fork VM id: 92
root@t2:/shared/kernel-fuzzer-for-xen-project# xl list
Name                ID    Mem VCPUs    State    Time(s)
Domain-0            0    8192     8    r----- 96534.3
ubuntu-20.04        86     0       1    --p---   11.9
(null)              92     0       1    --p---    0.0
root@t2:/shared/kernel-fuzzer-for-xen-project# cd ../vmtaint/
root@t2:/shared/vmtaint# xen-vmtrace 92 0 > vmtrace.log

```

```

root@t2:/shared/vmtaint# xl list
Name                ID    Mem VCPUs    State    Time(s)
Domain-0            0    8192     8    r----- 96524.2
ubuntu-20.04        86     0       1    --p---   11.9
(null)              91     0       1    --p---    0.0
root@t2:/shared/vmtaint# xl unpause 91
libxl: error: libxl_xshelp.c:201:libxl__xs_read_mandatory: xenstore read failed: '/libxl/91/type': No such file or directory
libxl: warning: libxl_dom.c:51:libxl__domain_type: unable to get domain type for domid=91, assuming HVM (reverse-i-search)`.': rm vmtrace^Cog
root@t2:/shared/vmtaint# ./vmtaint --help
./vmtaint: unrecognized option '--help'
Usage:
Step 1:
    --domid <domid>
    --save-state <file>

Step 2:
    --domid <domid>
    --load-state <file>

    --taint <address[:size]> (can be specified multiple times)
    --pt <file>
    --json <file>
    --skip-userspace (optional)
root@t2:/shared/vmtaint# ./vmtaint --save-state state.log --domid 86
Saving state
root@t2:/shared/vmtaint# 

```

```
root@t2:/shared/kernel-fuzzer-for-xen-project# forkvm 86
```

```
Fork VM id: 92
```

```
root@t2:/shared/kernel-fuzzer-for-xen-project# xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8192	8	r-----	96534.3
ubuntu-20.04	86	0	1	--p---	11.9
(null)	92	0	1	--p---	0.0

```
root@t2:/shared/kernel-fuzzer-for-xen-project# cd ../vmtaint/
```

```
root@t2:/shared/vmtaint# xen-vmtrace 92 0 > vmtrace.log
```

```
□
```

```
root@t2:/shared/vmtaint# xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8192	8	r-----	96524.2
ubuntu-20.04	86	0	1	--p---	11.9
(null)	91	0	1	--p---	0.0

```
root@t2:/shared/vmtaint# xl unpause 91
```

```
libxl: error: libxl_xshelp.c:201:libxl__xs_read_mandatory: xenstore read failed: '/libxl/91/type': No such file or directory
```

```
libxl: warning: libxl_dom.c:51:libxl__domain_type: unable to get domain type for domid=91, assuming HVM (reverse-i-search)`:': rm vmtrace^Cog
```

```
root@t2:/shared/vmtaint# ./vmtaint --help
```

```
./vmtaint: unrecognized option '--help'
```

```
Usage:
```

```
Step 1:
```

```
--domid <domid>  
--save-state <file>
```

```
Step 2:
```

```
--domid <domid>  
--load-state <file>  
  
--taint <address[:size]> (can be specified multiple times)  
--pt <file>  
--json <file>  
--skip-userspace (optional)
```

```
root@t2:/shared/vmtaint# ./vmtaint --save-state state.log --domid 86
```

```
Saving state
```

```
root@t2:/shared/vmtaint# xl unpause 92
```

```
libxl: error: libxl_xshelp.c:201:libxl__xs_read_mandatory: xenstore read failed: '/libxl/92/type': No such file or directory
```

```
libxl: warning: libxl_dom.c:51:libxl__domain_type: unable to get domain type for domid=92, assuming HVM
```

```
root@t2:/shared/vmtaint# xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8192	8	r-----	96538.0
ubuntu-20.04	86	0	1	--p---	11.9
(null)	92	38	1	r-----	1.4

```
root@t2:/shared/vmtaint# xl pause 92
```

```
root@t2:/shared/vmtaint# ./vmtaint --domid 86 --load-state state.log --taint 0xffffffffc02e0010:0x1000 --pt vmtrace.log --json /shared/cfg/5.4.0-48.json > taint.log
```



```

GNU nano 4.8                                taint.log
Tainting memory at 0xffffffffc02e0010 + 1000
fffffffffc02de0c5    movsx edi, byte ptr [rip + 0x1f44]
    Tainted reg: rdi: 0
fffffffffc02de0cc    call 0xffffffffc02de000
    Tainted reg: rdi: 0
fffffffffc02de000    nop dword ptr [rax + rax]
    Tainted reg: rdi: 0
fffffffffc02de005    mov edx, edi
    Tainted reg: rdi: 0
    Tainted reg: rdx: 0
fffffffffc02de007    push rbp
    Tainted reg: rdi: 0
    Tainted reg: rdx: 0
fffffffffc02de008    sar edx, 0x1f
    Tainted reg: rdx: 0
    Tainted reg: rdi: 0
fffffffffc02de00b    shr edx, 0x1e
    Tainted reg: rdx: 0
    Tainted reg: rdi: 0
fffffffffc02de00e    lea eax, [rdi + rdx]
    Tainted reg: rdx: 0
    Tainted reg: rax: 0
    Tainted reg: rdi: 0
fffffffffc02de011    mov rbp, rsp
    Tainted reg: rdx: 0
    Tainted reg: rax: 0
    Tainted reg: rdi: 0
fffffffffc02de014    and eax, 3
    Tainted reg: rdx: 0
    Tainted reg: rax: 0
    Tainted reg: rdi: 0
fffffffffc02de017    sub eax, edx
    Tainted reg: rdx: 0
    Tainted reg: rdi: 0
    Tainted reg: rax: 0

[ Read 90361 lines ]
^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify      ^C Cur Pos
^X Exit          ^R Read File    ^L Replace      ^U Paste Text   ^T To Spell     ^_ Go To Line

```

```

root@t2:/shared/vmtaint# xl list
Name                               ID   Mem VCPUs   State   Time(s)
Domain-0                           0  8192     8   r----- 96524.2
ubuntu-20.04                       86    0     1   --p---  11.9
(null)                             91    0     1   --p---   0.0

root@t2:/shared/vmtaint# xl unpause 91
libxl: error: libxl_xshelp.c:201:libxl__xs_read_mandatory: xenstore read failed: '/libxl/91/type': No such file or directory
libxl: warning: libxl_dom.c:51:libxl__domain_type: unable to get domain type for domid=91, assuming HVM (reverse-i-search)`:': rm vmtrace^Cog
root@t2:/shared/vmtaint# ./vmtaint --help
./vmtaint: unrecognized option '--help'
Usage:
Step 1:
    --domid <domid>
    --save-state <file>

Step 2:
    --domid <domid>
    --load-state <file>

    --taint <address[:size]> (can be specified multiple times)
    --pt <file>
    --json <file>
    --skip-userspace (optional)
root@t2:/shared/vmtaint# ./vmtaint --save-state state.log --domid 86
Saving state
root@t2:/shared/vmtaint# xl unpause 92
libxl: error: libxl_xshelp.c:201:libxl__xs_read_mandatory: xenstore read failed: '/libxl/92/type': No such file or directory
libxl: warning: libxl_dom.c:51:libxl__domain_type: unable to get domain type for domid=92, assuming HVM
root@t2:/shared/vmtaint# xl list
Name                               ID   Mem VCPUs   State   Time(s)
Domain-0                           0  8192     8   r----- 96538.0
ubuntu-20.04                       86    0     1   --p---  11.9
(null)                             92   38     1   r-----  1.4

root@t2:/shared/vmtaint# xl pause 92
root@t2:/shared/vmtaint# ./vmtaint --domid 86 --load-state state.log --taint 0xffffffffc02e0010:0x1000 --pt vmtrace.log --json /shared/cfg/5.4.0-48.json > taint.log

```

```

Tainted reg: rdx: 0
Tainted reg: rdi: 0
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rdi: 0
Tainted reg: rip: ffffffff02de01e
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rdi: 0
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rdi: 0
Tainted reg: rip: ffffffff02de023
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rax: 0
Tainted reg: rdi: 0
Tainted reg: rdx: 0
Tainted reg: rax: 0
Tainted reg: rdi: 0
Tainted reg: rip: ffffffff02de05e
Tainted reg: rdx: 0
Tainted reg: rdi: 1
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rdi: 1
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rdi: 1
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rdi: 1
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rdi: 1
Tainted reg: rax: 0
Tainted reg: rdx: 0
Tainted reg: rdi: 1
Tainted reg: rip: ffffffff02de041
Tainted reg: rax: 0
Tainted reg: rdx: 0

```

```
root@t2:/shared/vmtaint# xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8192	8	r-----	96524.2
ubuntu-20.04	86	0	1	--p---	11.9
(null)	91	0	1	--p---	0.0

```
root@t2:/shared/vmtaint# xl unpause 91
```

```
libxl: error: libxl_xshelp.c:201:libxl__xs_read_mandatory: xenstore read failed: '/libxl/91/type': No such file or directory
```

```
libxl: warning: libxl_dom.c:51:libxl__domain_type: unable to get domain type for domid=91, assuming HVM (reverse-i-search)`:': rm vmtrace^Cog
```

```
root@t2:/shared/vmtaint# ./vmtaint --help
```

```
./vmtaint: unrecognized option '--help'
```

```
Usage:
```

```
Step 1:
```

```
--domid <domid>
--save-state <file>
```

```
Step 2:
```

```
--domid <domid>
--load-state <file>

--taint <address[:size]> (can be specified multiple times)
--pt <file>
--json <file>
--skip-userspace (optional)
```

```
root@t2:/shared/vmtaint# ./vmtaint --save-state state.log --domid 86
```

```
Saving state
```

```
root@t2:/shared/vmtaint# xl unpause 92
```

```
libxl: error: libxl_xshelp.c:201:libxl__xs_read_mandatory: xenstore read failed: '/libxl/92/type': No such file or directory
```

```
libxl: warning: libxl_dom.c:51:libxl__domain_type: unable to get domain type for domid=92, assuming HVM
```

```
root@t2:/shared/vmtaint# xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8192	8	r-----	96538.0
ubuntu-20.04	86	0	1	--p---	11.9
(null)	92	38	1	r-----	1.4

```
root@t2:/shared/vmtaint# xl pause 92
```

```
root@t2:/shared/vmtaint# ./vmtaint --domid 86 --load-state state.log --taint 0xffffffff02e0010:0x1000 --pt vmtrace.log --json /shared/cfg/5.4.0-48.json > taint.log
```

```
]
```

Code released as open-source (MIT)

<https://github.com/intel/vmtaint>

vm transplantation

Getting some targets working in a Xen VM is difficult

No problem!

- VM forks need only CPU-state and memory
- Take snapshot on QEMU/KVM/Simics
- Load it up on Xen
- Fuzz away!

what's next

Automation

Capture system state using Intel[®] DCI

- Fuzz BIOS/SMM
- Fuzz bare-metal systems

Sandsifter ring0

Nested virt-support

things we didn't cover here

Fuzzing other OS's

Fuzzing Xen

Fuzzing userspace

Fuzzing black-box binaries

Fuzzing malware

thanks

Questions? Comments?

@tklengyel

Thanks goes to a whole lot of people who contributed & made this work possible!

Intel IAGS & CCG, Google ChromeOS team, Citrix, CERT.pl & the wider open-source security community releasing tools for everyone's benefit

<https://github.com/intel/kernel-fuzzer-for-xen-project>

<https://github.com/intel/vmtaint>